

Generative AI in architectural design: Application, data, and evaluation methods

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ABSTRACT

This paper presents a systematic review of generative artificial intelligence (AI) use in architectural design from 2014 to 2024, focusing on 1) AI models and theory-application gaps, 2) design phases, tasks, and objectives, 3) data types and contents, and 4) evaluation methods. Based on 161 journal papers selected using preferred reporting items for systematic reviews and meta-analysis (PRISMA), the analysis reveals the theory-application gap has been reduced by 96.09 %, from 62 to 2.5 years, highlighting rapid AI adoption since 2021 with generative adversarial networks (GANs) leading, and transformers and diffusion models gaining traction. For its application, AI is employed in schematic design phases in 68.94 %, while later phases remain underexplored. Regarding types of data used, images dominate at both input (52.8 %) and output (68.32 %), with multimodal and graph data showing promise. For evaluation, comparative evaluation was most utilized (60.9 %) supported by subjective assessment by authors (34.2 %) and third parties (17.4 %).

1. Introduction

Artificial intelligence (AI) is rapidly transforming architectural design. According to the Royal Institute of British Architects (RIBA) AI Report 2024 [1], 70 % of design firms now use AI, at least occasionally, during early design stage visualizations. While 57 % of respondents anticipate significant productivity gains from AI within two years, 35 % express concern that AI could threaten the profession. These contrasting views highlight gaps in understanding AI's capabilities in architectural design, underscoring the need for a clear assessment of how AI can support architect's tasks.

Advancements in AI, particularly neural networks (NNs), have enabled machines to learn intrinsic correlations between input and output pairs. The evolution from feedforward NNs (FFNNs) to advanced model architectures such as recurrent NNs (RNNs) and generative adversarial networks (GANs) has made it possible not only to recognize patterns in high-dimensional data such as images and languages but also to generate such data. Recent large AI models that are trained using massive datasets, such as transformers (e.g., generative pre-trained transformers (GPT) and LLaMA) and diffusion models (i.e., Stable Diffusion and DALL·E 2) can produce customized, high-dimensional outputs reflecting human intent.

These generative capabilities position AI as the next driver of

digitalization in architectural design, following the impact of computer-aided design (CAD), which successfully replaced conventional hand drawings by facilitating modification and replication. Building information modeling (BIM) has further shifted practice by supporting design integrity and offering a shared source for informed decision-making [2]. AI is expected to be the next frontier, supporting architects in interpreting diverse sources—including project scope, site conditions, user intents, and standard design documents—and efficiently exploring design alternatives while reflecting architects' multifaceted perspectives in making design decisions [3]. Consequently, there is a growing need to understand the status quo of these technologies' capabilities and their influence on design processes. To systematically examine the current landscape of AI in architectural design, this paper employs the preferred reporting items for systematic reviews and meta-analysis (PRISMA) 2020 guidelines [4]. Key questions emerge about how rapidly these technologies are integrated into architectural design, what design tasks they can currently support, and how to measure the performance of generative AI. To address these critical areas, this paper conducts a systematic review of academic literature to examine the state of AI adoption in architectural design. Specifically, this study seeks to address the following research questions:

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0926-5805/© 2025 Elsevier B.V. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

- 1) **How rapidly is AI being adopted in architectural design?** This question identifies the trends in the pace at which AI model architectures are introduced and utilized in architectural design to gauge how rapidly they are integrated into our domain. It informs how swiftly architectural education and training must adapt to address these changes, ensuring that individuals are equipped with both the soft and hard skills to leverage these advancements effectively [5]. In addition, it supports design firms in gauging the preparedness to adopt such advanced processes into their current practice, considering their resources including personnel and digital infrastructure [6].
- 2) **In which design phases and specific tasks is AI being applied, and which AI models are employed?** Identifying where AI is most applied within the design process and which specific models are utilized is essential for understanding how AI technologies influence the architectural design process. Investigating this question helps clarify which tasks AI can assist with or automate, offering insights into how architects can leverage these tools to enhance creativity, optimize design processes, or solve technical challenges. In addition, this facilitates researchers' understanding of the current gaps that need to be addressed, guiding future research opportunities.
- 3) **What are the data types and contents used in AI models for architectural design?** Architects manage various inputs—such as owner requirements, building codes, programming, conceptual images, and climate data—which are translated into design outputs such as drawings, 3D models, renderings, and text-based documents such as specifications and design descriptions [7]. Understanding the data types and contents handled by generative AI is essential for evaluating its potential and determining how it can seamlessly integrate into and improve existing design workflows.
- 4) **What are the methods and metrics used to evaluate AI models for architectural design?** Evaluating generative AI models is challenging because there is no single correct answer. In addition, the evaluation of architectural design requires assessing subjective criteria across multiple dimensions, such as novelty, coherence, semantic relevance, reliability, functionality, aesthetics, and overall design quality, which often rely on subjective criteria [8]. By identifying the evaluation methods, metrics, and performance indicators used in the previous studies, this research can offer guidelines for assessing AI tools, particularly those developed for architectural design.

By addressing these questions, this paper aims to contribute to the body of knowledge by providing a detailed understanding of the current state of architectural design in the era of generative AI. It serves as a guideline for architects, educators, and researchers to effectively leverage AI technologies to enhance design outcomes while understanding current limitations and future avenues for improvements.

The remainder of this paper is organized as follows: [Section 2](#) outlines the methods used to review the literature on generative AI in architectural design, following the PRISMA 2020 guidelines. [Sections 3, 4, 5, and 6](#) systematically review the AI models utilized for architectural design, the architectural design phases and tasks, data types and contents, and evaluation metrics, respectively. [Section 7](#) presents a detailed discussion of the findings and offers suggestions for future research opportunities, concluding the paper.

2. Methods

The process of the literature review in this study is described in [Fig. 1](#). First, studies on generative AI in architectural design were collected from the Web of Science (WoS) and Scopus databases. The collected research papers were screened according to the procedures in PRISMA 2020. We first conducted a brief bibliometric and keyword analysis, then systematically analyzed the 1) AI models utilized in architectural design and theory-application gap, 2) design phases, tasks,

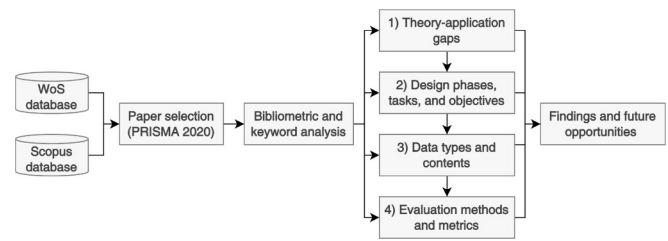


Fig. 1. Literature review process.

and objectives, 3) input/output data types and contents, and 4) evaluation methods. After the analyses of the trends in the four target topics, the results were analyzed in-depth to identify the current gaps and future opportunities.

2.1. PRISMA 2020 – eligibility criteria, search strategy, and selection process

This study followed PRISMA 2020 guidelines to conduct a systematic review. PRISMA is a tool designed to support researchers in conducting transparent systematic reviews or meta-analyses [4]. It provides a checklist with 27 items to follow when writing the manuscript's title, abstract, introduction, methods, results, and discussion sections. There are four types of systematic reviews classified by PRISMA: 1) new systematic reviews, 2) new systematic reviews, which include searches of databases, registers, and other sources, 3) updated systematic reviews, which include searches of databases and registers only, and 4) updated systematic reviews, which include searches of databases, registers, and other sources. We followed the second type: new systematic reviews that include searches of databases, registers, and other sources. From the papers identified in the databases, we further carried out a 'snowball' search to identify additional studies.

To search for papers relevant to the scope of this study, we employed the following search query to collect papers related to the generation of architectural design using AI: ('generative' OR 'generation' OR 'generate' OR 'generator' OR 'generated') AND ('artificial intelligence' OR 'AI' OR 'machine learning' OR 'deep learning') AND ('design' OR 'designer' OR 'designers' OR 'architecture' OR 'architect' OR 'architects'). The period was set from 01-Jan-2014 to 31-Jul-2024 to represent changes following the first paper on GANs in 2014 and to reflect 10 years. For keyword searching, this study restricted the document type to peer-reviewed journal papers to understand the current state of reputable and reliable sources.

We first identified studies from representative journal databases, including Web of Science (WoS) and Scopus. From an initial search across all research disciplines using the aforementioned keywords, we obtained 4846 documents from Scopus and 23,139 from WoS. However, many of these were irrelevant because the word "architecture" frequently refers to the structure of AI models, and "design" is often used to describe strategically formulating the AI model architecture to accomplish desired tasks or performance. Therefore, it was necessary to limit the disciplines. In the WoS search engine, we limited WoS categories to "Engineering Civil," "Construction Building Technology," and "Architecture." To filter the disciplines in the Scopus database, we first collected the journal names from Scimago Journal Rankings (SJR) and gathered all journals under the "Engineering" subject area, specifically "Civil and Infrastructural Engineering," "Building and Construction," and "Architecture" subject categories. Then we developed a Python code using `pybliometrics.scopus`, a library that supports application programming interface (API) based searching, to collect the papers from the Scopus database under relevant disciplines. In total, we searched papers from 552 journals identified as relevant research disciplines.

For the first step of screening, thoroughly reviewed the titles and abstracts of the collected papers and screened out papers that were far

from the roles of architects based on their titles or abstracts. Examples include papers dealing with topics such as transportation monitoring, traffic prediction, autonomous driving, digitizing existing buildings, monitoring indoor air quality or structural health of buildings, optimizing structural performance, predicting groundwater flow, and formulating construction materials. In the second step of screening, papers that were not able to be retrieved their full papers were removed.

In the third step, we assessed papers by reading full papers for eligibility. The four eligibility criteria employed in this study are 1) non-review papers, 2) inclusion of clear design outputs, 3) relevance to AIA design tasks, and 4) employment of NNs. Review papers and papers without clear design outputs were excluded because we aimed to collect original technical papers dealing with the generation of specific design results. Additionally, papers that were not relevant to the AIA design tasks or did not employ NNs were excluded in accordance with the scope of the review.

After screening papers from journal databases, we identified additional studies by conducting a reference search to collect influential publications—including journal as well as conference papers—commonly cited more than three times within the papers gathered from the prior step. Additionally, a citation search was conducted using Google Scholar, based on the relevance of titles of papers that cited the previously collected studies. Duplicate results from the original collections and publications outside the target time period were removed during the identification step. In the screening step, papers that could not be retrieved or failed to meet the eligibility criteria were excluded. Consequently, we compiled the total number of studies selected for the review.

2.2. Bibliometric and keyword analysis

In our review of the compiled papers, we conducted a bibliometric analysis to assess the number of publications within the target period, between 2014 and 2024, and their distribution across different journals. This analysis provides insights into the focus and impact of these publications within the academic landscape. Additionally, we performed a co-occurrence analysis of keywords extracted from the titles and abstracts of the reviewed papers using VOSviewer, a tool that generates co-occurrence networks and clusters keywords. We employed binary counting, meaning each combination of words in a paper was counted only once. To focus on the most relevant terms, we set a minimum occurrence threshold of five and filtered out 40 % of the less relevant words. Using the co-occurrence network generated by VOSviewer version 1.6.20, we analyzed each cluster by examining the included keywords to gain a comprehensive understanding of the trends in generative AI within architectural design.

2.3. Types of AI model architectures and theory-application gaps

In this analysis, we focus on identifying the AI model architectures adopted in architectural design and examining the pace at which they are being integrated into the field of architectural design. As previously mentioned, the term “architecture” can refer to both the structure of AI models and the field of architectural design. To avoid confusion, we will use the term “model architecture” to denote the structure of AI models.

We classified the collected papers according to their use of model architectures, considering both unmodified AI models and recent studies that address the architectural design domain while modifying AI model architectures. While model architectures are sometimes not mutually exclusive, it is possible to categorize them based on fundamental differences in training mechanisms. We categorized the papers into nine model architectures: FFNNs, RNNs, convolutional NNs (CNNs), GANs, encoder-decoder architectures, deep reinforcement learning (e.g., deep Q-learning, DQL), graph NNs (GNNs), transformers, and diffusion models.

FFNN: FFNNs are the simplest form of neural network architecture,

where information flows in one direction—from input to output—through multiple layers. Each neuron in a layer is connected to every neuron in the subsequent layer without any loops or cycles. Mathematically, the output of a single neuron can be expressed as Eq. (1):

$$f_{ij}^{(k)} = f\left(\sum_{i=1}^n w_i x_i + b\right) \quad (1)$$

where x_i represents the input features, w_i represents the weights associated with the inputs, b represents the bias term. $f(\cdot)$ represents the activation function (e.g., ReLU, Sigmoid), and y represents the output of the neuron. FFNNs are typically used for tasks such as classification and regression, where the relationships between input features and outputs are relatively straightforward, as seen in models such as multi-layer perceptron (MLP) and backpropagation NN (BPNN).

RNN: RNNs are designed to handle sequential data by maintaining a hidden state that captures information from previous time steps. This architecture is particularly useful for time series prediction, natural language processing, and other tasks where the order of inputs is important. The mathematical formulation for an RNN can be expressed as Eqs. (2) and (3):

$$h_t = f(W_h h_{t-1} + W_x x_t + b) \quad (2)$$

$$y_t = g(W_y h_t + c) \quad (3)$$

where h_t is the hidden state at time step t , x_t is the input at time step t , W_h , W_x , and W_y are weight matrices, b and c are bias terms, $f(\cdot)$ and $g(\cdot)$ are activation functions, and y_t is the output at time step t . Long short-term memory (LSTM) and gated recurrent unit (GRU) are popular variants of RNNs that address issues like vanishing gradients, enabling the model to capture long-range dependencies in sequences.

CNN: CNNs are particularly effective for tasks involving grid-like data, such as image processing. They use convolutional layers to extract features from input data, where each layer applies a filter to the input to produce a feature map. The output of a convolutional layer is given in Eq. (4):

$$f_{ij}^{(k)} = f\left(\sum_m \sum_n x_{i+m, j+n} \bullet w_{mn}^{(k)} + b^{(k)}\right) \quad (4)$$

where $f_{ij}^{(k)}$ is the output at position (i, j) in the k^{th} feature map, $x_{i+m, j+n}$ is the input at position $(i+m, j+n)$, $w_{mn}^{(k)}$ is the filter applied at position (m, n) in the k^{th} feature map, $b^{(k)}$ is the bias term and $f(\cdot)$ is the activation function. CNNs are commonly used for image classification, object detection, and segmentation tasks. Variants such as ResNet, VGG, and YOLO are widely employed for more complex image-processing tasks.

GNN: GNNs are specifically designed to operate on graph-structured data, and are therefore analyzed separately from other model architectures. The key operation in a GNN is the aggregation of information from a node's neighbors, which is then used to update the node's representation. A common formulation for a GNN layer as Eq. (5):

$$h_v^{(k+1)} = f\left(h_v^{(k)}, AGG\left(\left\{h_u^{(k)} : u \in N(v)\right\}\right)\right) \quad (5)$$

where $h_v^{(k)}$ is the representation of node v at layer k , $N(v)$ is the set of neighbors of node v , $AGG(\cdot)$ is an aggregation function (e.g., sum, mean, max), and $f(\cdot)$ is an update function. GNNs are used in applications such as property prediction and contextual similarity analysis.

GAN: GANs consist of two NNs—the generator and the discriminator—that compete against each other. The generator creates synthetic data samples, while the discriminator tries to distinguish between real and synthetic samples. The objective of the GAN is to solve the following min-max problem as Eq. (6):

$$\min_{G,D} \mathbb{E}_{x \sim p_{data}(x)} [\log D(x)] + \mathbb{E}_{z \sim p_z(z)} [\log(1 - D(G(z)))] \quad (6)$$

where $G(z)$ is the generator function that takes a noise vector z as input and generates a data sample, $D(x)$ is the discriminator function that outputs the probability of x being a real sample, $p_{data}(x)$ is the distribution of real data, and $p_z(z)$ is the distribution of the noise vector. GANs are primarily used to generate realistic images. Variants such as conditional GAN (CGAN) and Pix2Pix allow for more controlled results.

Encoder-decoder: Encoder-decoder are commonly used in tasks that require input transformation, such as machine translation and image segmentation. The encoder compresses the input into a lower-dimensional representation, while the decoder reconstructs the output from the input. The basic operation can be expressed as Eqs. (7) and (8):

$$h = \text{Encoder}(x) \quad (7)$$

$$y = \text{Decoder}(h) \quad (8)$$

where x is the input data, h is the latent representation, and y is the reconstructed output. U-net and autoencoders are examples of encoder-decoder architectures. Variants such as the variational autoencoder (VAE) introduce probabilistic elements to the latent space, allowing for more controlled data generation.

DQL: DQL is a type of deep reinforcement learning model that learns decision-making by approximating the Q-value function, which estimates the expected reward for taking a specific action in a given state. The Q-value function is iteratively updated using the Bellman equation, as shown in Eq. (9):

$$Q(s, a) = r + \gamma \max_{a'} Q(s', a') \quad (9)$$

where $Q(s, a)$ is the Q-value for state s and action a , r represents reward received after taking action a , γ is the discount factor, and s' is the next state. DQL models are particularly effective in environments that involve sequential decision-making, as they use this approach to determine optimal strategies. DQL has been successfully applied in parametric modeling, where it is used to optimize design processes by learning which actions lead to the best design outcomes over time.

Transformer: Transformers are based on self-attention mechanisms and have become the standard for tasks in natural language processing, such as translation and text generation. The key component of the transformer is the attention mechanism, which computes a weighted sum of values based on query-key similarity as expressed in Eq. (10):

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (10)$$

where Q (queries), K (keys), and V (values) are input matrices, d_k is the dimension of the keys, and the softmax function ensures that the attention weights sum to 1. Transformers are the basis for models of bidirectional encoder representations from transformers (BERT) and GPT that invoked the era of large language models (LLMs).

Diffusion Models: Diffusion models are a class of generative models that iteratively refine random noise into a structured output, such as an image. The process involves a forward diffusion process, where noise is added to data, and a reverse process, where the model learns to denoise the data. The objective can be expressed as Eq. (11):

$$\min_{\theta} \mathbb{E}_{t, x_0, \epsilon} [\| \epsilon - \epsilon_{\theta}(x_t, t) \|^2] \quad (11)$$

where x_0 is the original data sample, x_t is the noisy sample at time step t , ϵ is the noise added to the data, $\epsilon_{\theta}(x_t, t)$ is the model's prediction of the noise at time step t . Diffusion models such as Stable Diffusion and DALL-E 2 have been used for high-quality image generation reflecting the user prompts.

After classifying the collected papers into nine distinct model architectures, we first analyzed the number of publications using each

model architecture over the years, represented in a line graph. Then, we calculated the time lag between the first publication proposing each model architecture in the field of computer science and its earliest adoption in the literature addressing their use in architectural design, based on publication dates. This time lag is referred to as the theory-application gap in this study. We hypothesized that generative AI is being adopted in architectural design at an increasingly rapid pace. We quantitatively identified this by fitting the theory-application gaps of the model architectures to both an exponential decay equation and a linear regression model to evaluate which curve better represents the adoption trend. The exponential decay equation is defined as follows in Eq. (12):

$$y = a \times \exp(-b \times x) + c \quad (12)$$

where x represents a specific time, y represents the value on the exponential decay curve, a represents the initial time lag at the point of the model's introduction, b is the decay rate, indicating how quickly the time lag decreases, and c is the asymptotic value, representing the minimum time lag that the models eventually approach.

The linear regression model is defined as follows in Eq. (13):

$$y = d \times x + e \quad (13)$$

Where x represents a specific time, y represents the value on the exponential decay curve, d is the slope of the line, indicating the rate of change in the adoption gap over time, and e is the intercept, representing the initial adoption gap.

By comparing the actual trends against the fitted exponential and linear curves, we evaluated the pace of generative AI adoption in architectural design. The comparison was quantified using mean squared error (MSE) to determine which model better fit the observed data. When the trends aligned more closely with the exponential decay curve, it can be interpreted as indicating a gradual convergence toward a stable adoption rate. Conversely, when the trends were closer to the linear regression curve, it can be interpreted as suggesting a rapid and ongoing reduction in the adoption gap.

2.4. Design phases, tasks, and objectives

For the classification of design phases and tasks, we adopted the architect's services specified by B101™-2017 of the American Institute of Architects (AIA) [9]. The main architectural design phases in B101™-2017 are pre-design (PD), schematic design (SD), design development (DD), and construction documents (CD). Fig. 2 illustrates these design phases and tasks in each phase. The tasks in each phase are as follows:

In the PD phase, the architect analyzes the feasibility of the project (i.e., *feasibility study*), and plans adequate programs within the project's scope (i.e., *programming*). The focus here is on assessing the project's viability and establishing conditions for design without concentrating on scale-specific details.

During the SD phase, the architect explores design alternatives that best fit the project's scope (i.e., *design optioning*) and proposes sustainable design strategies (i.e., *environmentally responsible design consultation*). In this phase, the architect develops a basic concept for the project, considering factors such as site conditions, zoning requirements, and sustainability goals. The emphasis is on establishing the overall form and layout, providing a foundation for further development.

The DD phase involves refining the chosen design option to a higher level of detail. The architect produces more detailed drawings and specifications (i.e., *design development drawings*), including plans, elevations, and sections. They also engage in *user group meetings* to gather feedback, ensuring that the design meets the needs and expectations of the client and end-users.

In the CD phase, the architect prepares detailed drawings and specifications required for the project's construction. The *construction drawings* visually represent the integral and finalized systems of the

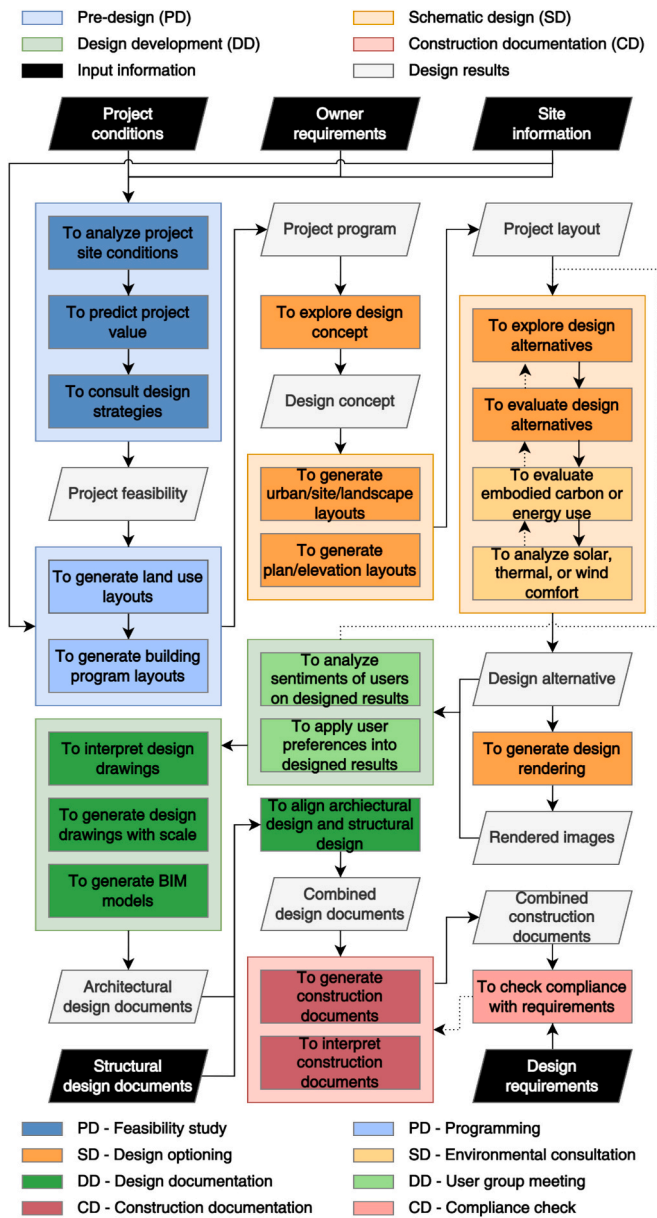


Fig. 2. Flow of design results across architectural design phases.

design, including structural details, electrical systems, plumbing, and finishes. These drawings provide a clear guide for contractors, ensuring that all physical aspects of the building are accurately constructed according to the design intent. *Specifications*, on the other hand, are written documents that outline the materials, products, and workmanship standards required for the project. While the drawings focus on the visual and technical elements, the specifications detail the quality, performance criteria, and installation procedures, ensuring that the construction meets the required standards. Together, these documents serve as a comprehensive set of instructions for the project to be built according to the architect's design.

While other services, such as design code review in the PD phase and cost estimations during the SD to CD phases, are also significant services provided by architects, they are not directly related to design generation and are considered separate tasks; therefore, they are not included in the scope of this review.

The analysis of design phases, tasks, and objectives was conducted in the following steps. First, we analyzed the trend in the number of publications per design phase over time using a line graph. Second, we

illustrated the distribution of tasks within each design phase using a stacked bar graph. Third, we presented a heatmap correlating design objectives with AI model architectures to demonstrate how AI is employed in architectural design. By extending the design phases and tasks based on the AIA Document B101™-2017, we classified the objectives of the reviewed papers, focusing on their target outcomes. When multiple AI model architectures are utilized to achieve a single design objective within a paper (e.g., through parallel or sequential combinations), weights were assigned to distribute their contributions proportionally. The weight w_i for each AI model i used for a single design objective was calculated as follows in Eq. (14):

$$w_i = \frac{1}{n} \quad (14)$$

where n represents the total number of AI models used for that objective. Finally, a detailed review was conducted for each identified objective to understand the specific AI models employed to achieve it.

2.5. Input-output data types and contents

In this analysis, we examine the data types and contents utilized by AI models in architectural design, spanning from PD to CD. This includes understanding the flow of design results and the associated data formats throughout each phase of the architectural workflow. The light gray parallelograms in Fig. 2 illustrate the design outcomes generated at each phase within the design process.

During the PD phase, architects gather extensive information, including project conditions, owner requirements, and site data. This information exists in various data types such as images, variables, and text. Architects analyze project site conditions, predict project values, and develop design strategies based on this diverse information. Feasibility analyses determine the overall usage and proportion of land and building areas through programming, often represented using bubble diagrams in attributed graph formats that reflect the sizes and topology between different programmatic elements.

As the project advances to the SD phase, architects explore various design concepts that may involve different data types, including abstract concept images, 3D massing models, or even musical elements. Once design concepts are established, architects integrate the program within the concept, forming a coherent layout within the site and building. The project layout is typically represented as drawings, allowing architects to adjust compositions and explore alternative designs. Providing environmentally responsible solutions has become a common task, considering factors such as embodied carbon, energy use, and analyses of solar, thermal, or wind comfort. These results are usually depicted as images with simulated overlays and variables indicating performance metrics. Additionally, generating detailed design renderings is an important task in the SD phase.

During the DD phase, a significant task involves conducting user group meetings to gather feedback on design preferences. Based on user feedback, architects may undertake reworks and explore alternative design options. Once a design option is finalized, it is technically represented as design documents, typically in the form of 2D drawings and increasingly as BIM models. By aligning these documents, including interdisciplinary elements such as structural and mechanical, electrical, and plumbing (MEP) systems, architects combine and elaborate the design documents to reflect construction-level details in the CD phase.

In the CD phase, architects generate comprehensive construction documents that include detailed specifications necessary for project completion. Compliance with regulatory requirements and client expectations is meticulously checked to ensure that all design specifications meet the necessary standards.

To align with the scope of this review, we categorized the data types that AI models receive as input and produce as output in each study. There are five distinct data types—image, variable, text, graph, and 3D

model—and four multimodal combinations identified from previous studies. By “variable”, we refer to a structured alphanumeric data type, while, by “text,” we refer to an unstructured alphanumeric data type. Additionally, specific data contents were classified as detailed in Section 6. For example, if a proposed method integrates parametric modeling to generate a 3D building model using AI-generated data, the input- data type is image. Similarly, if CAD drawings are converted into images before being processed by an AI model, the image is recorded as the input data format.

To understand the overall distribution of input-output data types and content used in previous studies, a Sankey diagram mapping the connections between input data types, input content, output content, and output data types was created and analyzed. Additionally, heatmaps are generated to correlate these data formats with AI architectures and design objectives, providing insights into how different data types are processed by various AI model architectures and how they influence the application of AI in architectural design. In this step, the weighting method introduced in Section 2.4. is applied to account for the multiple use of AI model architectures within a single paper.

2.6. Evaluation methods, types, and metrics

To understand how previous studies assessed the performance and effectiveness of AI models, we classified their evaluation methods into four distinct categories: comparative evaluation, assessment by the authors, assessment by the third party, and machine-based assessment. While comparative evaluation involves comparing AI outputs and references such as datasets, expert designs, or simulations, assessments—whether human or machine—evaluate the design results directly, without necessarily involving a ground truth or benchmark. We further classified the evaluation methods used for each type as follows:

Comparative evaluation compares AI-generated results with predefined references, allowing for a direct performance assessment. *Dataset validation* uses a portion of the training dataset, commonly an 80/20 split, to validate model outputs, ensuring consistency with known ground truth. Some studies compare AI-generated designs with expert-generated designs (i.e., *expert results*), focusing on areas like efficiency or performance. Similarly, other studies compare AI-generated results with *simulated results*, such as Monte Carlo simulations [10], finite element modeling (FEM) [11], and physics-based engines such as energy, radiance, light [12], and computational fluid dynamics (CFD) simulations [13]. These comparisons assess how closely AI can replicate such simulated results with reduced computational costs or without the need for engines (i.e., surrogate models).

Assessment by authors involves the authors themselves evaluating the AI-generated designs, without comparison to external benchmarks. *Visual interpretation* relies on the authors' judgment to assess the quality of the design, often without metrics. *Numeric interpretation* involves calculating values such as proportions or distances to quantitatively assess aspects like design features such as floor-area ratio (FAR) and building coverage ratio (BCR) [14]. *Criteria-based interpretation* determines whether the design satisfies specific project requirements, such as regulatory compliance.

Assessment by third parties such as evaluations from the public or domain experts, is based on external feedback on AI-generated designs. *Surveys* collect feedback from the public or students, primarily focusing on preferences or usability. *Expert evaluations* involve experienced architects and engineers assessing the practicality of AI designs and determining their real-world viability. The *Turing test* evaluates the human-likeness of AI outputs by asking evaluators to distinguish between human- and AI-generated designs, providing insights into the AI's ability to replicate human creativity.

Automated assessment by machines involves machine-based evaluations of AI-generated outputs using algorithms or models. *Algorithm-based assessments* use rule-based systems such as DepthmapX to compute the visibility score of rooms [15], or algorithms such as graph

editing distance (GED) [16] to evaluate the topological similarity between input and output data. *Supervised* evaluations rely on trained models, such as support vector classifiers (SVCs) trained for classifying synthetic images [17], CNNs, and vision transformers trained for rating the quality of rendering [18]. *Unsupervised* assessments evaluate design results by themselves. For instance, the contrastive language-image pre-training (CLIP) score [19] compares the similarity between embeddings of input descriptions and output results to assess how well the generated results conform to the given prompt. Similarly, the Inception score [20] is used to evaluate the diversity of data generated by AI models.

Previous studies used these evaluation methods and types individually and collectively. To quantify their use across different studies, an UpSet plot is utilized to illustrate the frequency and overlap of various evaluation methods. The number of papers using each method and type is also analyzed. Additionally, the distribution of these evaluation types across input-output data types is presented as a normalized heatmap, providing a comprehensive view of the assessment landscape of AI-driven architectural design. This step also considered the weights for multiple use of AI model architectures within a single paper. Furthermore, frequently used evaluation metrics and performance indicators are identified, and their applications are explained.

3. Bibliometric and keyword analysis

Fig. 3 depicts the flow of identification, screening, and inclusion of studies for review following the PRISMA 2020. All the outputs of the screening procedures were validated by the authors to ensure accuracy and eliminate errors.

From the keywords search, we collected 666 papers from WoS and 299 papers from Scopus. After removing 103 duplicate papers by comparing the digital object identifiers (DOIs), we had 792 papers for screening. Of these, 519 papers were excluded during the title and abstract screening in the first step, and 1 paper was excluded in the second step due to the full text being unavailable. In the third step, 179 papers were excluded, including 17 review papers, 6 that lacked specific design outputs, 122 irrelevant to AIA design tasks, and 34 that did not employ NNs. Consequently, 93 papers were selected from journal databases.

In addition to 93 papers, we collected 230 more papers through reference search (75 papers) and citation search (155 papers). A total of 200 papers remained, after removing 19 duplicates that had already been collected from the previous step and 11 papers outside the time scope of the review (i.e., those published before 2014). One paper was excluded in the first screening step for being unretrievable. In the eligibility assessment step, 131 papers were excluded for the following reasons: 19 were review papers, 8 lacked specific design outputs, 37 were irrelevant to AIA design tasks, and 67 did not employ NNs. Ultimately, 68 papers were selected from the additional search, increasing the total number of studies included in the review to 161. Detailed information of all 161 selected papers, including their publication years, titles, authors, journal/conference names, and DOIs, is provided in Appendix A as Supplementary Material.

3.1. Bibliometric analysis

Fig. 4 shows the historical trend of generative AI publications in architectural design. From 2014 to 2018, there were only a small number of research papers published. However, after 2019, there is a noticeable upward trend in the publication of generative AI-based research in architectural design, with a sharp increase in 2023. As of July 2024, the number of published papers stands at 53, which is comparable to the total of 59 for the previous year, even though only a little more than half of the year has passed.

A total of 60 journals and 7 conferences have published research on generative AI in architectural design. Table 1 summarizes the journal sources of the reviewed papers. The top ten journals account for 56.52 % of all papers. *Automation in Construction* has published the most research

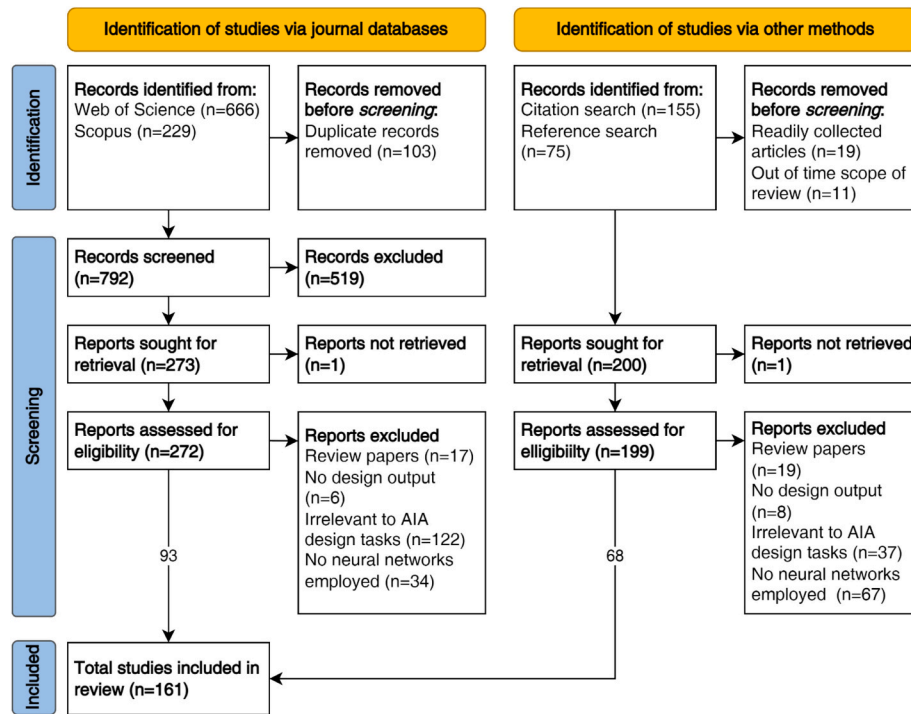


Fig. 3. PRISMA 2020 flow diagram for systematic reviews.

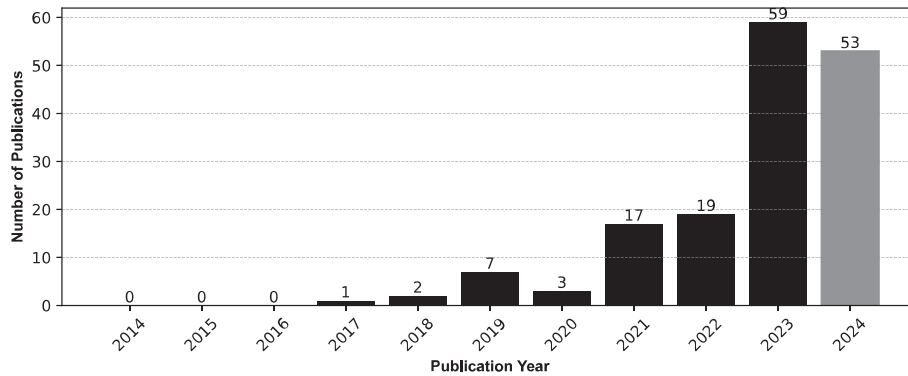


Fig. 4. Number of published papers from Jan. 2014 to July 2024.

papers (23 papers), followed by *Buildings* (15 papers), the *International Journal of Architectural Computing* (12 papers), and the *Journal of Building Engineering* (12 papers). Additionally, 39 out of the 60 journals (65 %) have published only one paper on generative AI in architectural design, indicating that the topic is still in its nascent stage. Seven conference papers were published in different conferences not only from computer-aided design domain but also from those from computer vision domains (Table 2).

3.2. Keywords analysis

From the title and abstract fields of the screened papers, 4467 keywords were identified using VOSviewer. After removing keywords that occurred fewer than five times, 179 keywords remained. Following a filtering process that removed 40 % of the less relevant words, a total of 107 keywords were retained. The resulting co-occurrence network of keywords is illustrated in Fig. 5.

The three clusters were quantitatively analyzed with a focus on occurrences and link strengths, resulting in the following insights:

Cluster 1: Parameter-based Architectural Design Integration

(Red): Cluster 1 is the largest cluster, comprising 39 keywords and a total link strength of 2999, resulting in a normalized link strength of 76.90. The central keywords “value” (23 occurrences, 167 link strength) and “parameter” (23 occurrences, 160 link strength). This highlights the emphasis on optimizing design parameters and outcomes through the integration of AI technologies into digital design methods, such as parametric modeling and BIM. Additional keywords, such as “prediction” (17 occurrences, 144 link strength) and “simulation” (15 occurrences, 111 link strength) further illustrate the diverse applications of AI in predictive modeling and design simulations. These terms are connected to others such as “surrogate model” (6 occurrences, 32 link strength), “engineering” (8 occurrences, 91 link strength), “early stage” (6 occurrences, 45 link strength), and “early design stage” (9 occurrences, 73 link strength), indicating that the current application of AI in architectural design is focused on the timely evaluation of design outcomes and predicting optimal design parameters from the early stages of the design process.

Cluster 2: Ability of Artificial Intelligence in the Field of Architectural Design (Green): Cluster 2 includes 34 keywords and exhibits a total link strength of 2863, with a normalized link strength of 84.21. The

Table 1
Journal sources of reviewed papers.

Journal name	Number of publications
Automation in Construction	23
Buildings	15
International Journal of Architectural Computing	12
Journal of Building Engineering	12
Building and Environment	6
Engineering Structures	6
Computer-Aided Civil and Infrastructure Engineering	6
Energy and Buildings	4
Journal of Digital Landscape Architecture	4
Advanced Engineering Informatics	3
Architectural Science Review	3
Environment and Planning B: Urban Analytics and City Science	3
Technology Architecture + Design	3
Applied Artificial Intelligence	2
Engineering, Construction and Architectural Management	2
Expert Systems with Applications	2
Frontiers of Architectural Research	2
Journal of Computational Design and Engineering	2
Land	2
Sustainability	2
Sustainable Cities and Society	2
ACM Transactions on Graphics	1
Architectural Design	1
Architectural Engineering and Design Management	1
Architectural Intelligence	1
Architecture and Planning Journal (APJ)	1
Asian Journal of Civil Engineering	1
Building Research & Information	1
Building Simulation	1
Cities	1
Computer Graphics Forum	1
Computers in Industry	1
Construction Innovation	1
Developments in the Built Environment	1
Digital Government: Research and Practice	1
Earthquake Engineering & Structural Dynamics	1
EGA Revista de Expresión Gráfica Arquitectónica	1
Frontiers in Built Environment	1
Frontiers in Ecology and Evolution	1
Frontiers in Psychology	1
Heliyon	1
IEEE Transactions on Visualization and Computer Graphics	1
Intelligent Buildings International	1
International Journal on Document Analysis and Recognition (IJDAR)	1
Journal of Asian Architecture and Building Engineering	1
Journal of Building Performance Simulation	1
Journal of Computational Design	1
Journal of Computing in Civil Engineering	1
Journal of Construction Engineering and Management	1
Journal of the International Association for Shell and Spatial Structures	1
Knowledge and Information Systems	1
Landscape Architecture Frontiers	1
Nexus Network Journal	1
Open House International	1
Results in Engineering	1
Rita: Revista Indexada de Textos Académicos	1
Scientific Reports	1
Smart and Sustainable Built Environment	1
Technologies	1
Total	154

keyword “field” dominates this cluster with 22 occurrences and 199 link strength, emphasizing the significance of domain-specific AI applications. Notably, “development” (21 occurrences, 183 link strength), “architect” (19 occurrences, 131 link strength), “creativity” (20 occurrences, 145 link strength), and “efficiency” (15 occurrences, 138 link strength) collectively reflect a focus on advancing and refining AI applications to increase the efficiency of creative architect’s design tasks. Keywords such as “technology” (15 occurrences, 120 link strength) and

Table 2
Conference sources of reviewed papers.

Conference name	Year
Proceedings of 15th International Conference on Machine Vision Applications (MVA)	2017
Proceedings of 38th Annual Conference of the Association for Computer Aided Design in Architecture (ACADIA)	2018
Proceedings of 1st International Conference on Computational Design and Robotic Fabrication (CDRF)	2019
Proceedings of 25th Computer-Aided Architectural Design Research in Asia (CAADRIA) Conference	2020
Proceedings of 16th European Conference on Computer Vision (ECCV)	2020
Proceedings of 2021 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)	2021
Proceedings of 2021 IEEE/CVF International Conference on Computer Vision (ICCV)	2021

“ability” (15 occurrences, 112 link strength) suggest an exploration of AI’s creative capabilities and technological advancements within architectural contexts. Keywords such as “drawing” (20 occurrences, 154 link strength), “text” (16 occurrences, 128 link strength), “idea” (12 occurrences, 114 link strength), “image data” (6 occurrences, 55 link strength), and “image generation” (5 occurrences, 55 link strength) illustrate that the motivation for employing AI in architectural design lies in the ability of AI models to translate human ideas into design outcomes through mediums such as text or images.

Cluster 3: Challenges and Practical Implementation (Blue): Cluster 3, characterized by 34 keywords, has a total link strength of 2814 and a normalized link strength of 82.76. The keyword “intelligence” is the most prominent, with 46 occurrences and a link strength of 318, followed by “quality” (25 occurrences, 170 link strength) and “challenge” (21 occurrences, 155 link strength). These keywords highlight difficulties encountered in AI implementation to attain quality output. Other significant keywords include “design method” (20 occurrences, 143 link strength) and “workflow” (15 occurrences, 117 link strength), all of which emphasize the ongoing efforts to enhance design processes and streamline architectural workflows using AI. The terms “potential” (16 occurrences, 116 link strength) and “effectiveness” (16 occurrences, 110 link strength) suggest that while AI has the potential to be effectively adopted in the architectural design process, the current focus is on exploring and fully harnessing this potential.

4. AI model architectures and theory-application gaps

The analysis of the publication trends from 2014 to 2024 for various AI model architectures in architectural design reveals a significant shift in adoption patterns (Fig. 6). Some studies employed multiple AI model architectures, consequently the total number of publications exceeded the original count of 161 papers. Initially, there were no publications related to these AI models in architectural design until 2017, when CNNs appeared with a single publication proposing a method to interpret design drawings. In 2018, two studies were conducted using GANs, marking the beginning of the integration of generative AI into the field of architectural design. The adoption of GANs continued to grow, reaching 27 publications in 2023, highlighting its prominence in applications in architectural design (81 papers, 50.31 %). Other model architectures such as FFNNs, CNNs, and encoder-decoder models began to see increased adoption starting in 2021, with FFNNs and CNNs (25 papers, 15.53 % each) showing notable growth. The data also indicates the emergence of newer model architectures such as transformers, with nine papers (5.59 %), and diffusion models, with 23 papers (14.29 %). Although introduced later, both have seen rapid uptake, particularly in 2023 and 2024, highlighting the importance of model architecture that supports designers by generating results aligned with the design intent. The overall trend suggests that the adoption of AI models in architectural design has increased significantly in recent years, with GANs leading the way, followed by a broader integration of other prior and

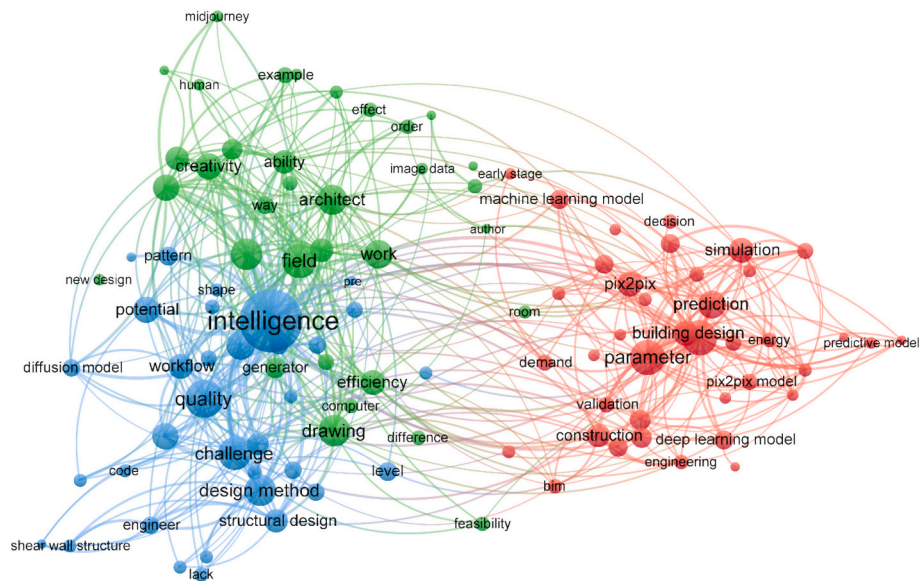


Fig. 5. Co-occurrence network of keywords from screened papers.

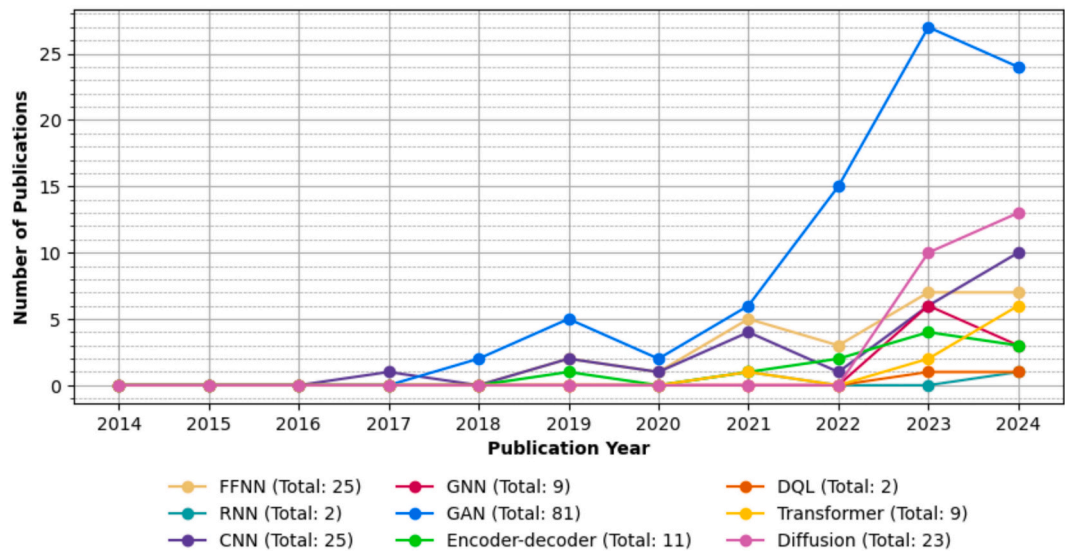


Fig. 6. Number of publications per AI model architecture.

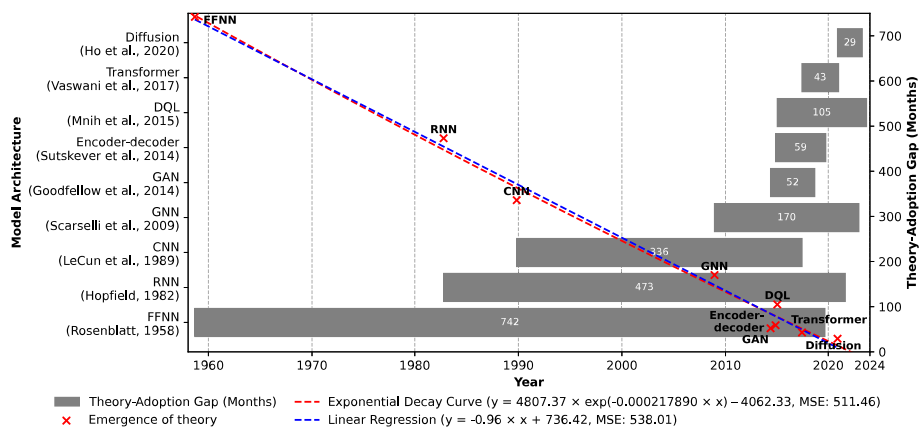


Fig. 7. Theory-application gap of AI model architectures in architectural design (Fitted exponential decay curve: $y = 4807.37 \times \exp(-0.000217890 \times x) - 4062.33$, Linear regression: $y = -0.96 \times x + 736.42$).

advanced AI model architectures. This rapid increase in the number of publications underscores the growing interest and reliance on AI technologies in the architectural design generation.

Fig. 7 illustrates the decreasing theory-application gaps as newer AI models are adopted in architectural design. Early models such as FFNNs, introduced in 1958, took over 61 years to be applied, while RNNs and CNNs had gaps of approximately 40 and 28 years, respectively. More recent models, such as GNNs and GANs, experienced significantly shorter lags, with GANs being adopted just four and a half years after their introduction. This trend continues with models such as transformers and diffusion models, which saw applications within about three and a half years and two and a half years, respectively. This pattern highlights the accelerating pace at which AI technologies are being integrated into architectural design.

The fitted exponential decay curve ($y = 4807.37 \times \exp. (-0.000217890 \times x) - 4062.33$) and linear regression both represent a rapid reduction in theory-application gap ($y = -0.96 \times x + 736.42$), with exponential decay curve slightly better representing the trend with MSE at 511.46 compared to MSE at 538.01 for linear regression. For the fitted exponential decay curve, the parameter $a = 4807.37$ was expected to represent the initial time lag when these models were first introduced, indicating a significant delay in their early adoption within architectural design. However, this exceedingly large value does not logically correspond to the actual initial gaps observed in the data. Similarly, the decay rate $b = 0.000217890$ is extremely small, suggesting an imperceptibly slow rate of decrease in the time lag, which contradicts the observed acceleration in adopting newer AI models. The asymptotic value $c = -4062.33$, theoretically representing the minimum time lag if the trend continued indefinitely, was negative—an illogical result implying that AI models would be applied before their introduction. Despite its slightly lower MSE, these discrepancies highlight that the exponential decay model has limitations in its interpretability and logical consistency. In contrast, the linear regression model offers a simpler and more interpretable trend, indicating a consistent and steady reduction in the theory-application gap over time. The slope $d = -0.96$ represents a uniform rate of decrease in the adoption gap, and the intercept $e = 736.42$ corresponds to the initial adoption gap. While the MSE of 538.01 is slightly higher than that of the exponential decay model, the linear regression model avoids the logical inconsistencies seen in the exponential decay parameters, making it a more reliable representation of the observed data.

The comparison between the two models demonstrates that while the exponential decay model provides a slightly better numerical fit, the linear regression model better captures the consistent narrowing of the gap and aligns with the observed trend without introducing unrealistic assumptions. This finding suggests that the adoption of generative AI in architectural design is progressing steadily, with the gap narrowing at a relatively constant rate and no indication of slowing down. As advancements in AI continue to accelerate and interdisciplinary collaboration becomes more prevalent, it is anticipated that the adoption of new AI technologies in architectural design will occur with increasing immediacy. In other words, the field is expected to witness a continued narrowing of the theory-application gap.

5. Design phases, tasks, and objectives

Extending the design phases and tasks based on Document B101™–2017 of AIA, we further classified the objectives of the previous studies focusing on the target results upon review of the screened papers. Table 3 outlines the number of publications and references for each design objective as identified in this study. The details of the objectives identified through the review are elaborated in subsection 5.3.

5.1. Trends in generative AI use by the design phase

Fig. 8 shows the trend in the number of publications by design phase

Table 3
Design phases, tasks, and objectives.

Phase	Task	Objective	Number of publications
Pre-Design (PD)	Feasibility Study	To predict project value.	1
		To consult design strategies.	2
		To analyze project site conditions.	3
	Programming	To generate land use layouts.	3
Schematic Design (SD)	Design Optioning	To generate building program layouts.	3
		To explore design concepts.	21
		To generate urban/site/landscape layouts.	13
		To generate plan/elevation layouts.	28
		To generate design rendering.	10
		To explore design alternatives.	8
		To evaluate design alternatives.	4
		To evaluate embodied carbon or energy use	13
		To analyze solar, thermal, or wind comfort.	14
		To generate design drawings with scale.	1
Design Development (DD)	Design Development Drawings	To interpret design drawings.	4
		To generate BIM models.	6
		To align architectural design and structural design.	20
	User Group Meetings	To analyze the sentiments of users on designed results.	2
		To apply user preferences to designed results.	2
		To interpret construction drawings.	2
Construction Documents (CD)	Construction Specifications	To check compliance with standard specifications.	2
			1

and year. Among the design phases, the SD phase is the most dominant, particularly from 2021 onward, accounting for 68.94 % of the total with 111 among 161 papers. This highlights a strong focus on using AI during the early stages of design, where conceptual exploration is crucial. AI applications in the DD phase are also growing, with an increase to 13 publications in 2024, accounting for 21.74 % with 35 papers in total. This suggests an expansion beyond initial concepts to more detailed stages of development. While a dozen studies have been conducted in the PD phase, particularly with a rise in 2023, it accounts for 7.45 %. The CD phase applications have seen minimal research, with three papers accounting for 1.86 %, indicating that AI's application in this phase remains underdeveloped. Overall, the rapid growth in publications, especially in the SD and DD phases, reflects the maturing use of AI in architectural design, while the limited focus on the PD and CD phases

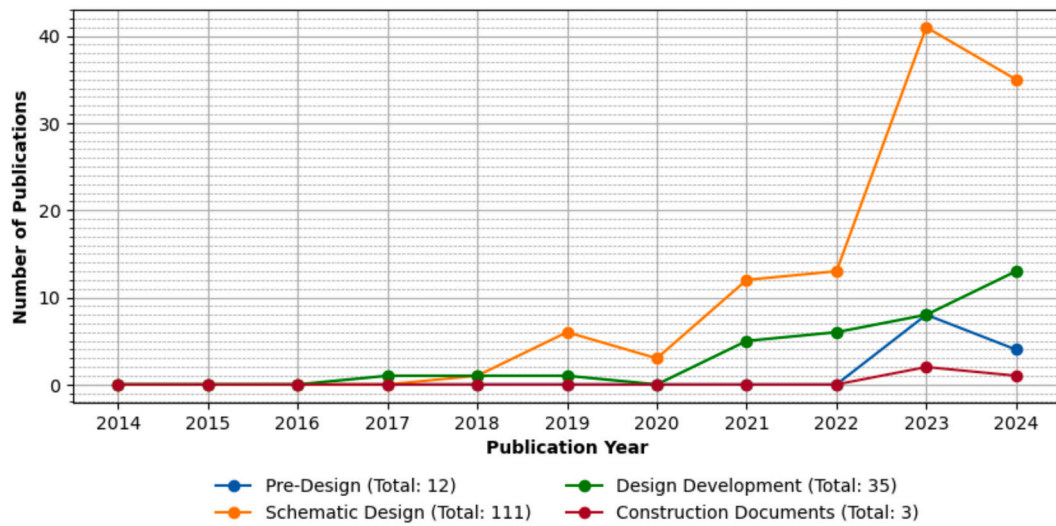


Fig. 8. Number of publications per design phase and year.

highlights areas for potential future research.

5.2. Trends in generative AI use by design task

Fig. 9 illustrates which task in each phase generative AI has been applied. In the PD phase, AI is primarily used for *feasibility studies* and *programming*, with six papers each. In the SD phase, most cases are concentrated on *design optioning*, with 84 papers, followed by *environmentally responsible design consultation*, with 27 papers. During the DD phase, generative AI is predominantly used in *design development drawings*, with 31 papers, and to a lesser extent in *user group meetings*, with 4 papers. The CD phase shows limited generative AI use, with two papers focusing on *construction drawings* and one focusing on *construction specifications*. Overall, the data suggests that AI is mostly applied in the SD and DD phases, particularly for cases related to *design optioning* (52.17 %), cases related to *environmentally responsible design consultation* (16.77 %), and *design development drawings* (19.25 %), occupying 88.20 %.

5.3. Trends in generative AI use by design objective

The heatmap in Fig. 10 shows how different AI model architectures are used for various architectural design tasks. It highlights the frequency with which each AI model is applied to specific design objectives. To account for cases where papers use multiple AI models, weights were assigned, and the weighted totals are shown in parentheses. The following papers provide detailed descriptions of how each AI model is

applied to different design objectives.

5.3.1. PD - feasibility study

To predict project value (one paper): This objective involves assessing the financial viability of the project by estimating potential costs, revenues, and overall economic impact. Jiang et al. [21] proposed a method to automatically evaluate land values by predicting urban maps with land use, built form, social factors, site context, and building use masked based on the images of urban layouts using GANs. Their application emphasizes the potential of image-based processing to incorporate holistic urban factors affecting the value of the project site.

To consult design strategies (two papers): The emergence of LLMs was enabled by training transformer-based language models using a massive amount of natural language data ranging from technical documentation to scholarly articles. These LLMs, trained on diverse and extensive datasets, have demonstrated the capacity to provide viable design strategies upon designers' requests [22]. Shelby and da Silva [23] developed a question-answering system for landscape architects using a BERT-based retrieval-augmented generation (RAG) method, which refers to external data sources for text generation [24]. George et al. [25] attested to the capacity of GPT for recommending suitable plants for landscape design based on ecological, stylistic, functional, and aesthetic parameters. These advancements highlight the growing role of AI-driven systems in assisting architects with data-informed, contextually relevant design strategies.

To analyze project site conditions (three papers): Analyzing

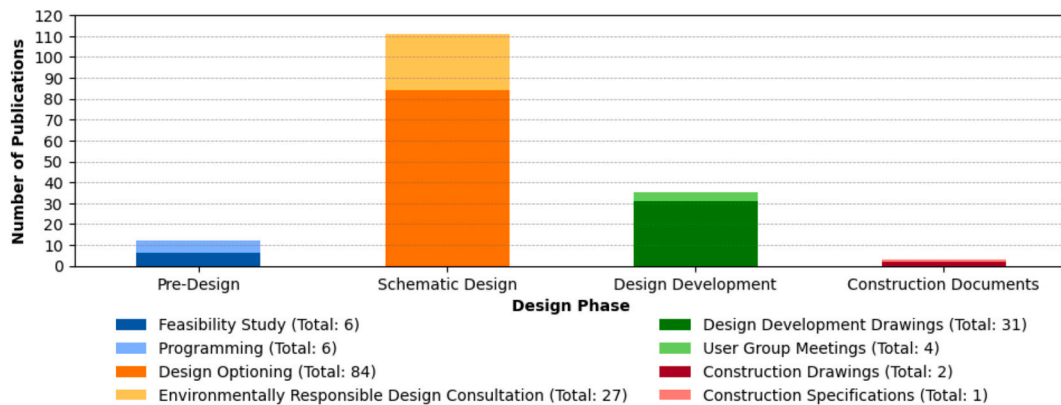


Fig. 9. Number of publications per design task in each phase.

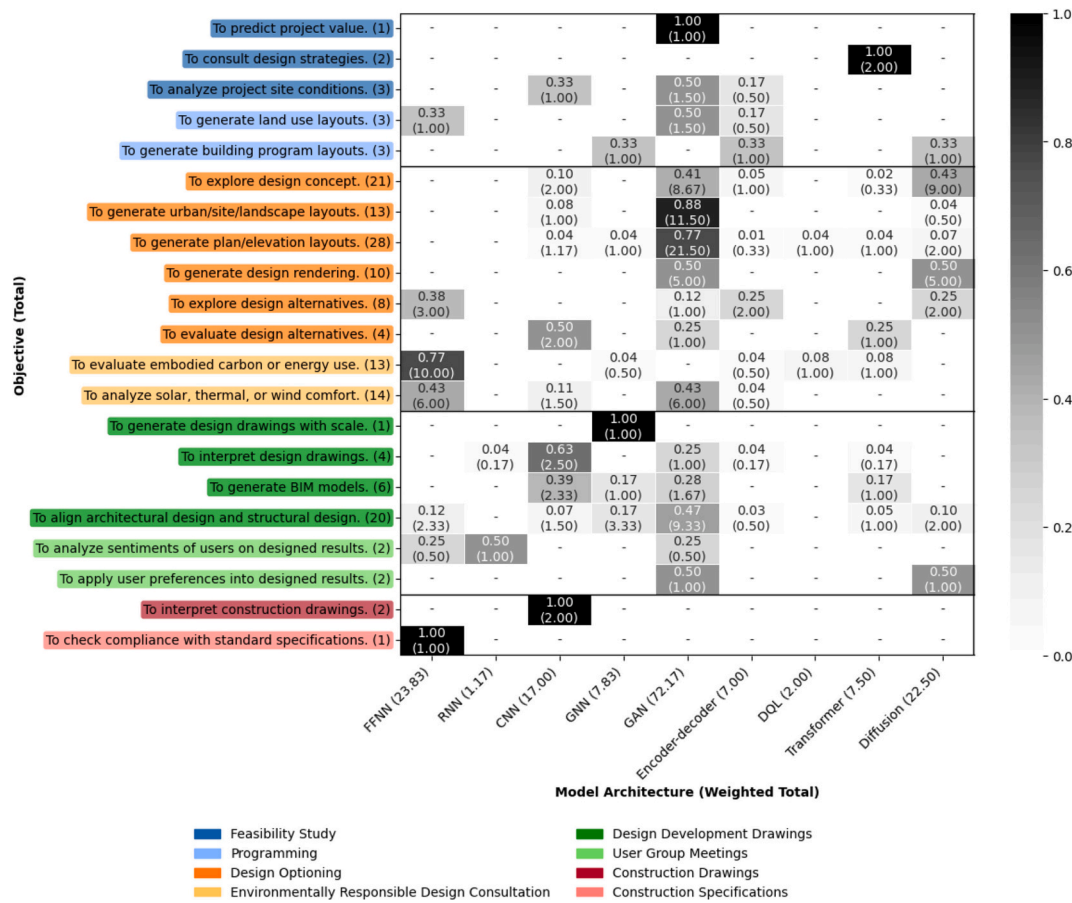


Fig. 10. AI model architectures deployed for various design objectives.

project site conditions such as topology and urban context of the adjacent buildings is a crucial task in the early stages of a project. He [26] proposed a method to automatically retrieve CAD drawings of the project site using a light convolutional encoder-decoder and compared it to a U-net-based approach. The AI was trained to segment types of land and elevation from aerial images. Similarly, Mayrhofer-Hufnagi and Ennemoser [27] proposed a method to predict the elevation of buildings and site topology, enabling automated analysis of the orientation of the urban layout. Nakhaee and Paydar [28] developed a method to predict urban radiation from 360° street images and visualize the results using augmented reality for preemptive site analysis. These studies commonly processed AI-generated images coupled with parametric tools to generate topography and geometries of surrounding buildings.

5.3.2. PD - programming

To generate land use layouts (three papers): Allocating the functions of the lands is a significant task for optimizing urban or site use. Wan et al. [29] trained various GANs and encoder-decoder models to recommend land use based on surrounding land use patterns following the human-guided green plant coverage. Park et al. [14] proposed a method using a modified Pix2Pix model to predict optimal land use in urban districts. They simultaneously predicted land use, FAR, and BCR by generating a red, blue, and green (RGB) image, with each channel representing one of the design conditions. Liu et al. [30] developed a method employing an FFNN and space syntax to predict energy-efficient commercial site configuration, considering the distances between the facilities.

To generate building program layouts (three papers): Generating building program layouts involves creating room configurations as a bubble diagram, which illustrates the dimensions and connections of

spaces. Sun et al. [31] trained a VAE to generate spatial bubble diagrams on building footprints. Similarly, Su et al. [32] developed a method to generate a spatial graph using a discrete denoising diffusion model (D3M) based on an image of a building footprint. Park et al. [33] developed a system that recommends possible spatial graphs from a spatial graph database using SimGNN. These proposed methods for programming commonly aim to support architects in making timely, purpose-oriented spatial configurations during the PD phase.

5.3.3. SD - design optioning

To explore design concepts (21 papers): Ideating design concepts is a fundamental objective in the SD phase, involving the creative process of refining design concepts to form the basis of architectural design. It was among the most extensively focused on objectives along with aligning the architectural design and structural design. Two studies focused on generating and improving the quality of concept sketches using GANs. Akdoğan and Balaban [34] trained a CycleGAN to generate conceptual spatial sketches from images of microorganisms, while Qian et al. [35,36] proposed a method to enhance the quality of concept sketches for high-rise building exteriors by combining Self-Sparse GAN with VGG 16 and 19. Several studies also explored generating 3D concept mass through various approaches, such as creating a mesh from sketches [37] or 2D images [38,39], producing voxel masses based on geometric operations [40], and completing point cloud populations to form a concept mass from featured point clouds [41]. Additionally, a study proposed generating a voxel-encoded image from random noises and reconstructing it as a conceptual spatial mesh [42]. Similarly, Newton [43] proposed a data augmentation-based method to train GANs to generate images that reflect specific styles (e.g., architect,

historical period, and city) using noise-based training data. **Methods for generating concept images from prompts, primarily using diffusion models, were extensively introduced, with a focus on aligning the output image with the project's text description, including objectives [44], concepts [45], emotions [46], conditions [47–50], and styles [51–53].** Başarır et al. [54] further proposed a method to generate conceptual building images based on regional music by embedding local songs into audio spectrograms and coupling them with local building images to train a Pix2Pix model. These studies commonly aim to use AI tools to support the ideation of initial design concepts.

To generate urban/site/landscape layouts (13 papers): Generating urban, site, and landscape layouts involves the detailed arrangement of entities within a given land area. This objective differs from land use layout generation in the PD phase, where the focus is on determining the use of the land, whereas here, the emphasis is on arranging buildings within an urban district [55], allocating buildings on a project site [56], and designing landscape masterplans [57]. Various GANs were primarily utilized for this objective, including Pix2Pix, CycleGAN [55], and GauGAN [58]. Additionally, domain-specific GANs such as context-aware site planning generative adversarial networks (CAIN-GAN) [59], site-embedded generative adversarial networks (SEGAN) [60], and Urban-GAN [61] were proposed. Other AI architectures included an input-controlled spatial attention U-Net (ICSA U-Net), a CNN, for on-site building allocation [62], and a diffusion model to enhance the resolution of the generated landscape image [62]. **Since these models typically generate image-based results, many studies integrated parametric modeling to translate these results into 3D models [55].**

To generate plan/elevation layouts (28 papers): Plan layouts refer to the arrangement of rooms and spaces within a floor plan and represent the most extensively explored objectives. While generating a plan layout shares similarities with building programming layouts in determining optimal spatial configurations, it goes further by specifying the shape of each space and organizing their overall context. Extensive studies focused on generating building plan layouts for different building types and their unique characteristics, including houses [63–66], high-rise residential buildings [67,68], residential complexes [69,70], public buildings [71], museums [70,72], school classrooms [73], and hospital emergency rooms [74]. Most of these studies used GANs to generate spatial layouts within defined boundaries [63,75,76]. Luo and Huang proposed a GAN-based method to utilize room schedules to generate a preliminary layout for human elaboration [76]. Other approaches incorporated partial layouts along with boundaries to capture architects' intentions in the design process, leveraging diffusion models [77,78]. Another area of focus was generating layouts from graph-formatted bubble diagrams [64,65], which expanded to combining bubble diagrams with building footprints as an image [67,79], as well as multimodal inputs combining graphs and images [17,66]. While these methods primarily used GANs, diffusion models, and GNNs, Veloso and Krishnamurti [70] proposed a method using multi-agent DQL to generate layouts for various project types (i.e., house, museum, and residential complex) based on the site conditions represented on an image. There were also studies on generating elevation layouts given the exterior profile using CGAN [80] and Pix2Pix [81]. In summary, these studies primarily focused on generating project-specific layouts by utilizing patterns learned from similar projects, enabling AI models to produce designs that align with established architectural norms and requirements.

To generate design rendering (ten papers): Design rendering involves creating detailed visual representations of proposed designs, aiming to produce realistic images or models that effectively convey the design intent to clients and stakeholders. Several studies have explored generating rendered images from line drawings, such as CAD [82] and sketch masterplans [83], as well as sketches of building perspectives [84,85]. Studies have also proposed generating photorealistic exterior renderings from building mass perspectives captured in BIM models

[18] and creating interior videos from interior views within BIM models [19]. Both approaches utilized diffusion models, incorporating BIM-generated perspectives and input prompts as multimodal inputs for customized renderings. Diffusion models were also used to reflect user preferences as an image output such as generating 2D rendering assets [86] and applying instructed styles for interior rendering [87,88].

To explore design alternatives (eight papers): Exploring design alternatives is essential for diversifying options while staying aligned with the overall design concept. Unlike exploring design concepts, this step involves refining existing ideas, such as parametric models or visualized building designs, which can be adjusted or further developed. For instance, Zheng et al. [89] proposed a method to predict input design parameters for optimal structural performance and constructability of shell-structured street furniture, given subdivision rules and boundary conditions. Similarly, Aymen et al. [90] developed a method to adjust the design parameters of building components (e.g., walls, slabs, and windows) based on conditions like land length, width, setbacks, FAR, and the number of building blocks. Zheng and Yuan [91] introduced a technique for exploring design alternatives for skyscrapers by adjusting control points of horizontal profiles, based on the stylistic preferences of renowned architects (or their design firms) such as Le Corbusier, MAD, Gensler, Zaha Hadid Architects, Tadao Ando, BIG, and SOM. These studies utilized FFNN, a relatively traditional AI model architecture, to predict optimal design parameters for parametric rules, given design conditions as input [89]. Danhaive and Mueller [92] proposed a method to automatically explore the structural design space using a conditional VAE (CVAE), incorporating surrogate models to explore latent space for structural optimization. Similarly, Mirra and Pugnale [93] proposed a method utilizing boundary profiles and controlling variables to generate depth maps and reconstruct a populated surface for a shell structure using VAE. These methods focus on integrating AI's data generation capabilities and applying them to create input data for parametric models, resulting in more robust, parameterized outputs that can be seamlessly incorporated into conventional computer-based design processes, as opposed to relying solely on AI-generated images and variables. Another significant use case of generative AI for exploring design alternatives is modifying existing design images based on user prompts. Kim [94] proposed a method to alter the style of architectural perspectives using StyleGAN along with CLIP, trained on descriptions and images from existing building projects, in response to user-provided keywords. With the rise of diffusion models, applying specific styles to existing architectural images through prompts became possible without fine-tuning [95]. Hanafy [95] examined the functions supported by diffusion-based platforms, such as DALL-E 2, Midjourney, and Stable Diffusion, including partial modification, resizing, and upscaling, and proposed potential use cases for architectural image generation and modification. Similar to design rendering generations, these approaches highlight the importance of incorporating user preferences when exploring design alternatives.

To evaluate design alternatives (four papers): Once design alternatives have been generated, the next objective is to evaluate them based on specific criteria. Nourkojouri et al. [96] trained a Pix2Pix model to predict evacuation congestion from building layouts to evaluate fire emergency evacuations in the early design stage. Moradi et al. [97] proposed a method for automatically evaluating building plan layouts generated using the parametric design by extracting spatial adjacency graphs through room and door segmentation using various CNNs, including YOLOv4, EfficientNet, YOLOv5, DetectorRS, and RetinaNet. Other studies focused on using AI to evaluate the visual aesthetics of design images. Park et al. [98] proposed a method to automatically evaluate the aesthetics of facade modules using three indexes ranging from 1) ugly to beautiful, 2) ambiguous to unambiguous, and 3) boring to interesting, utilizing MobileNet. Gui et al. [88] developed a visual transformer to augment the parametric modeling-based rendering process in underground spaces and optimize visual comfort using reinforcement learning. These methods are commonly focused on employing

AI models to assess and select the optimal alternatives, considering both subjective and objective factors.

5.3.4. SD - environmentally responsible design consultation

To evaluate embodied carbon or energy use (13 papers): An important aspect of environmentally responsible design is assessing the environmental impact of a design in terms of embodied carbon and energy consumption [99], as well as predicting cooling and heating capacity [100] and the power generation capacity of rooftop solar energy systems [101]. These approaches mainly employed **parametric modeling** (e.g., grasshopper in McNeel Rhino) and simulation algorithms to collect the predicted energy use and embodied carbon based on changes in design parameters and conditions. Then, AI models, primarily FFNNs, were trained to predict the estimated results without running costly simulations (i.e., surrogate models). These surrogate models were further extended by being coupled with optimization functions such as the multi-objective ant lion optimization algorithm (MOALO) [102] and the nondominated sorting genetic algorithm II (NSGA-II) [100], as well as by using the predicted results to evaluate rewards in reinforcement learning [99] and deep deterministic policy gradient (DDPG) [103].

To analyze solar, thermal, or wind comfort (14 papers): Similar to evaluating embodied carbon or energy use, analyzing environmental factors of design involves using predicted results from simulations as datasets to train surrogate models. These factors include daylight autonomy (DA) and its variations, illuminance [104], sun hours [105,106], radiation [107], visual comfort [108], as well as wind flow [109]. Several studies also expanded this analysis into optimization by integrating the predicted outcomes with NSGA-II [105,108,110]. Compared to models for embodied carbon or energy evaluations, which often rely on structured data, surrogate models for analyzing environmental factors like solar, thermal, or wind comfort tended to favor GANs and CNNs, using color-mapped images of simulated results for training.

5.3.5. DD - design development drawings

To generate design drawings with scale (one paper): While several previous studies generated images visually similar to the drawings yet without dimensions (e.g., floorplan effect) [73], only one study proposed a method to generate a floor plan with annotated dimensions of building components. Babakhani [111] proposed a three-step method in which the first step, k-means clustering was used to process user input and generate initial room allocations. In the second step, a GNN trained on thousands of architectural models refined the spatial dimensions and relationships. In the last step, a genetic algorithm optimized the placement of rooms and structural elements on the Cartesian plane, ensuring topological relations, geometrical constraints, and user-defined requirements were satisfied. This highlights the potential of integrating GNNs with parametric modeling to represent topological and dimensional semantics, enabling better spatial relationships and design parameter representation in building design.

To interpret design drawings (four papers): Interpreting design drawings involves extracting and understanding the semantics from architectural drawings, with CNNs mostly employed for this task due to their capacity to detect and extract semantic information such as text, symbols, and building components. Dodge et al. [112] proposed a three-step method for interpreting floor plans that utilizes fully convolutional networks (FCN) for wall segmentation, Faster R-CNN for detecting building elements such as doors, sliding doors, kitchen appliances, bathtubs, sinks, and toilets, and optical character recognition (OCR) for extracting metadata. Pizarro et al. [113] collected high-resolution images of residential buildings annotated with walls and slabs to address data scarcity in this field. They proposed a method using U-net for automated wall segmentation from architectural drawings. Schönfelder et al. [114] introduced a two-step text recognition method involving object detection and OCR. They also introduced a dataset of plans with annotated text and conducted a comparative evaluation of object

detection architectures (Tesseract, EAST, CRAFT, Faster R-CNN, YOLOv5, YOLOR, YOLOv7, and YOLOv8) and OCR tools (PARSeq, MATRN, EasyOCR, and Tesseract), revealing that YOLOv7 and PARSeq performed best. There was also a study utilizing GANs for drawing interpretation. Huang and Zheng [115] proposed using Pix2PixHD to segment room types on floor plans and applied the same approach to enhance the visual representation of the room types.

To generate BIM models (six papers): While the requirement to include BIM models as submittals is increasing, the industry still faces a hurdle in adopting full BIM, a process where a primitive BIM model is consecutively updated throughout the project life cycle. Accordingly, CAD-to-BIM is still a dominant process that requires significant efforts in representing readily described project semantics. Several studies have focused on reducing such efforts by automatically recognizing the building components and associated design parameters for BIM model generation. Zhao et al. [116] proposed a method utilizing Faster R-CNN and YOLO to automatically detect grid heads, columns, and beams in architectural drawings. They employed PyTesseract, a Python OCR library, to extract design parameters and generate industry foundation classes (IFC) instances. Similarly, Urbietta et al. [117] developed an IFC-based method to segment building components such as walls, columns, beams, slabs, doors, and windows using Mask R-CNN, combined with OCR via Google's Cloud Vision, from both architectural and structural drawings to generate a comprehensive BIM model. Li et al. [118] took a different approach by focusing on full BIM readiness, proposing a method for automating the generation of layouts for precast high-rise residential buildings using standardized household BIM modules. The approach integrates graph convolutional neural networks (GCNNs) to follow the IFC precast scheme, enabling efficient spatial arrangement and optimization of precast layouts for enhanced design accuracy and performance. Ghannad and Lee [119] proposed a coupled GAN (CoGAN)-based method to generate plan layouts, which are then adjusted using a parametric algorithm to vectorize them for modularized parametric BIM generation. For plans generated by CoGAN, junctions are identified through pixel-wise analysis, and spaces are classified using shape grammar and semantic analysis. Finally, IFC processing is applied to create BIM models. Kim et al. [120] proposed an automated method for placing doors in floor plan layouts, addressing the less-explored area of door placement. The method used ResNet-50 to predict door positions from labeled images, followed by Pix2Pix-GAN to generate layouts with doors. The study also introduced a parametric algorithm for generating doors in BIM models by identifying walls from the mask image and placing doors based on calculated coordinates. There was also a study focused on reducing the technical barriers to BIM adoption by leveraging natural language understanding and generation capabilities of transformer models, specifically LLMs. Jang et al. [22] proposed a framework for integrating LLMs with BIM systems as a conversational interface. The LLM chaining framework, named natural-language-based architectural detailing through interaction with AI (NADIA), interprets user intent and generates BIM instances in a textual format, adhering to object-oriented programming (OOP) principles. These text-based outputs are then converted into the BIM environment via API scripts to generate and modify BIM instances.

To align architectural design and structural design (20 papers): Aligning architectural and structural design is the most extensively studied objective, representing 57.14 % of the DD phase objectives and 12.42 % of all screened papers. Although more extensive studies have specifically employed generative AI in structural design [121], we screened papers that specified their objectives as aligning structural building components with existing architectural designs, excluding those focused on enhancing structural design itself. Various AI model architectures employed in the reviewed studies including FFNNs [122], CNNs [123], GNNs [122], GANs [123], encoder-decoder models [124], transformer models [125], and diffusion models [126,127], highlight the extensive research conducted in generating and aligning structural elements with architectural designs. Most studies focused on segmenting

potential shear structures in architectural plans, covering not only shear walls [123], but also non-structural walls, slabs, beams, columns, and windows and doors [123,128,129]. Other studies focused on automatically designing the structural layout of frame structures, particularly in placing beams and columns over architectural layouts [130]. Additionally, Zhang et al. [122] developed a domain-specific GAN, StrucTopo-GAN, which was augmented with MLP and GNN to extract wall and beam edges, and column and non-column nodes from architectural plans. Similarly, Köseenciğ [131] proposed a Pix2Pix model for structural grid generation, augmented with rule-based adjustments. These studies are commonly aimed to enable the automatic generation of structural topology for further adjustment. In summary, extensive research efforts reflect the significance of collaboration in the DD phase, demonstrating the potential of generative AI in streamlining this collaborative process which requires in-depth domain understanding.

5.3.6. DD - user group meetings

To analyze the sentiments of users on designed results (two papers): Analyzing user sentiments involves gathering and evaluating feedback from end-users on the proposed design. This objective is crucial for understanding how the design is perceived and whether it meets user needs. Chang et al. [132] developed a method to analyze user responses to buildings and classify sentiment into binary classes—positive and negative—using an LSTM network. The authors further extended this research by analyzing electroencephalography (EEG) data to predict binary user sentiment on displayed design results [133], utilizing an MLP trained on an EEG dataset augmented with GAN. These efforts highlight the future potential for AI-driven, real-time feedback in user group meetings, enabling more responsive and personalized design processes.

To apply user preferences to designed results (two papers): Applying user preferences involves customizing design outcomes based on the specific needs and desires of users. While this objective is similar to modifying design results based on designers' prompts, the following studies specifically mention their aims of incorporating public preferences into architect-designed projects. Quan [134] proposed a framework that integrates public participation into the urban design process, enabling users to provide preferred urban case images to modify an existing urban plan using a deep convolutional GAN (DCGAN). Guridi et al. [135] introduced a method for incorporating user feedback into the design of public parks using Dream Studio, a diffusion model-based platform. Although these studies are in their early stages and lack significant technical differences from methods used for reflecting designers' prompts, they underscore the importance and potential of end-user participation in urban planning, made possible through the integration of generative AIs into the design process.

5.3.7. CD - construction drawings

To interpret construction drawings (two papers): Both studies on the interpretation of construction drawings focused on material quantity takeoff. Jamieson et al. [136] concentrated on understanding MEP drawings, proposing a method to detect symbols (e.g., valves, pumps, meters) using YOLOv7 and Faster R-CNN. Chowdhury and Moon [137] proposed a method to detect unit objects of concrete structures from different views of a BIM model and automate the quantity take-off using Mask R-CNN and Tesseract, an RNN-based OCR. Both studies commonly focused on automatically and accurately obtaining the respective amounts of building components for project construction by utilizing robust object detection through CNN-based models.

5.3.8. CD - construction specifications

To check compliance with standard specifications (one paper): Ensuring construction specifications align with national standards is crucial to prevent unexpected design changes and contractual risks. Lee et al. [138] proposed an FFNN-based method that uses a bag-of-gram embedding to automatically reference relevant national

standards in construction specifications. This study highlights the potential of generative AI to streamline the design review process, reducing both time and manual effort.

6. Input-output data types and contents

Upon analyzing the input and output data types and contents, we observed a diverse range of content being entered as input and returned as output. Table 4 outlines the data types and contents identified in the previous studies. A total of 30 unique contents have been identified across these studies, using five distinct data types: images, variables, text, graphs, and 3D models.

Images serve as a 2D visual representation of designs, composed of distinct cells (i.e., pixels), and are widely used as both input and output in AI-driven architectural design. Boundaries define the limits of the design space [63,66], while layouts depict the arrangement of buildings, spaces, or components [77]. Bubble diagrams are used to illustrate spatial relationships between different spaces [31], and technical drawings provide detailed illustrations of building components, including plans, elevations, and sections [112]. Conceptual inputs, such as sketches [37] and concept images [34,39], guide AI in refining or generating designs. Additionally, images such as aerial views [26], perspective views [135], 3D model views [18], and 360° perspectives [28] help convey the broader context of urban environments and building features. For output, AI models can use prediction overlays, which apply gradient colors to represent numerical values on design images [112,115]. AI-generated photorealistic renderings [18], rendering materials [86], and videos [19] enhance the visual comprehension of anticipated design outcomes.

Variables are used to represent design features and constraints in architectural design. Design parameters (e.g., dimensions, distances, and quantities) are often used as input variables in parametric modeling to define and control key aspects of a design [12,139]. These parameters allow for flexible adjustments, enabling the generation of multiple design variations based on changes in these values. They also serve as outputs, reflecting the optimized design parameter predicted by AI models. Additionally, conditions, such as environmental factors or building code restrictions, serve as constraints that influence the design process [76,127]. On the output side, AI models generate predicted values, including metrics like annual energy consumption or recommended land use, which are forecasted based on design outcomes [92]. One study also used EEG data, captured while viewing design results, as an input sequential variable [133].

Text is primarily used to convey descriptive design intentions to AI, influencing the design outcomes as an input type. Prompts guide AI models in performing design tasks [22,23], while user responses provide valuable feedback on the proposed designs [132]. Additionally, key documents, such as specifications, which detail design decisions and their implementation in construction, are processed in text format to ensure consistency in standards and methods [138]. For outputs, AI generates text-based answers in response to user prompts [23,25], and data structures filled with the properties used to represent BIM objects, adhering to object-oriented programming principles of BIM software [22].

Graphs represent the relationships between design elements in architectural design, using nodes and edges to capture complex contextual relationships within a structured format. As both input and output, graphs can depict layouts, illustrating the arrangement of spaces or elements within a design [64]. Additionally, bubble diagrams in graph form show spatial relationships between rooms or zones, allowing AI models to process and optimize space organization [65,66]. Drawings as graphs illustrate spatial or design elements within a project [140,141].

3D models provide a comprehensive method for digitally representing architectural designs in spatial formats, offering an intuitive understanding of design outcomes. As both input and output, 3D point

Table 4

Types and contents of input and output data for generative AI in architectural design.

Type	Input/ Output	Content	Description
Image	Input/ Output	Boundary	A defined border that limits the design space, often concealed using colors.
		Drawing	A technical illustration of buildings, including plans, elevations, and sections.
		Layout	The arrangement of buildings, spaces, or building components within a design.
		Perspective view	A representation of the design from a specific viewpoint.
		Sketch	A freehand drawing conceptualizing design ideas or drafts.
		Aerial view	An overhead view of an urban area or project site.
		3D model view	A captured image from a 3D modeling tool showcasing the design.
		Concept image	An image visually conveys the initial idea or concept of a design.
		360° perspective view	A panoramic image that provides a 360-degree view of a scene.
	Input	Façade module	A unit or section of a building's façade.
		Audio spectrogram	A visual representation of sound frequencies from local music.
		Random noise	A visual depiction of randomly generated noise patterns.
		Prediction overlays	An image with gradient colors representing numerical values overlaid on the original design.
		Rendering	A photorealistic visual representation of a design.
		Rendering asset	An image asset that is utilized in the rendering process.
		Bubble diagram	A diagram illustrating spatial relationships between different spaces using bubbles.
		Video	A rendered video of the interior design results.
		Encoded voxel	A visual representation of volumetric cells (i.e., voxels).
	Variable	Design parameters	Numerical or categorical variables defining design features such as dimensions, distances, and quantities, are often used in parametric modeling.
		Conditions	Constraints or factors such as environmental conditions or building codes influence the design process.
		Electroencephalography (EEG)	A captured brain activity data.
	Input	Predicted values	Numerical or categorical values forecasted by a model, are often based on design outcomes.
	Output		
	Text	Specification	A detailed document outlining design decisions and their application in construction, establishing consistent standards and methods.

Table 4 (continued)

Type	Input/ Output	Content	Description
	Input	Prompt	Instructions or questions guiding AI models in completing a design task.
		User response	Feedback or reactions provided by users regarding a design.
	Output	Answer	A text-based response to a user prompt.
		Data structure and values	A text representation of BIM objects, following a data scheme based on object-oriented programming classes and properties.
	Graph	Bubble diagram	A graph illustrating spatial relationships between spaces.
		Drawing	A graph representation of spatial or design elements within a project.
		Layout	A graph representation of the arrangement of spaces or elements in a design.
	3D model	3D point clouds	A collection of data points that capture the geometric shape of a design in 3D space.
		Concept mesh	A network of segmented faces that define the shape of a design in 3D space.
		Concept voxel	A volumetric representation of a design using cells in 3D space.

clouds capture a design's geometry using thousands of data points in 3D space [41]. On the output side, AI-generated designs may be represented as concept voxels, using volumetric cells to define the design's shape in 3D space [40,43], or as concept meshes, where segmented faces come together to form a complete 3D structure [37,38].

In some cases, data in the same format, such as design parameters and conditions, are combined as input variables [142]. Additionally, multimodal inputs involve using data in different formats together, such as images combined with text [18,135], variables [59,76,119], or graphs [17,66], as well as graphs paired with variables [99,111]. Some studies also feature multimodal outputs, where both images and variables are returned together [116,117].

6.1. Distribution of input-output data types and contents

The analysis of data types and contents processed by AI models for architectural design reveals a diverse range of input and output data types and contents (Fig. 11).

Image is the most extensively used input data type (85 papers, 52.8 %), accommodating a wide range of input content such as boundaries (27 papers, 16.77 %), drawings (24 papers, 14.91 %), layouts (22 papers, 13.66 %), and other content. Variables are used for input contents such as design parameters (11 papers, 6.83 %), conditions (eight papers, 4.97 %), and their combinations (six papers, 3.72 %). Text is primarily utilized for user prompts (16 papers, 9.94 %), with additional instances for user responses and specifications (one paper each). The graph is used to represent contextual relationships between components of layouts, drawings (two papers each), and spatial bubble diagrams (four papers). There is also a study that used 3D point clouds as input [41]. In addition, multimodal formats were used in 21 papers (13.04 %), where input content was often represented through combinations of images with text, variables, or graphs, underscoring the complexity of the architectural design process, which requires input data to reflect nuanced details. For example, images of drawings, layouts, and boundaries were accompanied by design parameters and conditions. Similarly, images of drawings, 3D models, and perspective views were often paired with

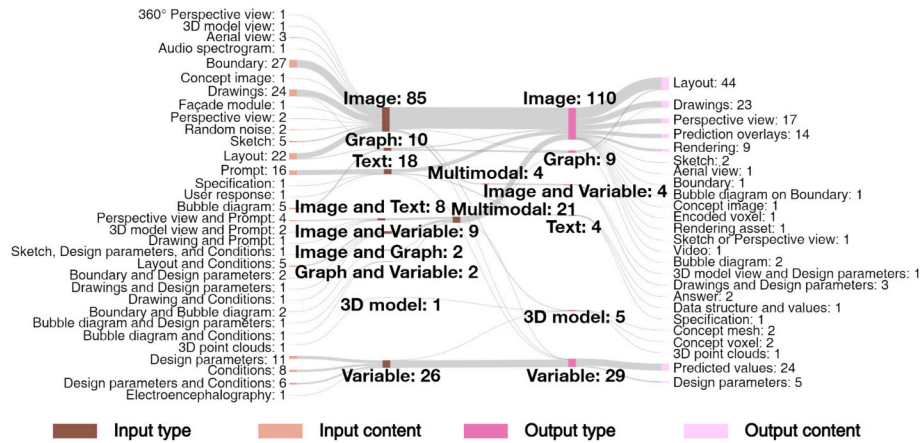


Fig. 11. Flow of input-to-output data type and content using AI in architectural design.

prompts specifying required changes. Other combinations included embedding building footprints with bubble diagrams (image with graph) [17,66], or bubble diagrams with associated design parameters and project conditions (graph with variable) [99,111].

Image also dominates among output data types (110 papers, 68.32 %), representing content such as layouts (39 papers, 24.22 %), drawings (21 papers, 13.04 %), and perspectives (17 papers, 10.56 %). These outputs are critical in conveying the visual and spatial aspects of architectural designs. Beyond these standard visual outputs, images also capture specialized content, including prediction overlays (14 papers), renderings (nine papers), and other content. Variables are also frequently used output data type (29 papers, 18.01 %), particularly as predicted values (24 papers, 14.91 %) or design parameters (five papers). Text outputs, while less frequent, are used for providing answers to users' questions (two papers), referencing standard documents, and generating text-formatted BIM objects (one paper each). The graph is specialized in representing design results with contextual semantics such as layouts (five papers), the relationship between building components on the drawings, and bubble diagrams (two papers each). Lastly, 3D models are generated to visualize the concept massing using meshes [37,38], voxels [40,43], and 3D point clouds [41]. Four studies return an image and variables (multimodal output), as drawings with detected objects and design parameters (three papers) and a 3D model view with detected objects and design parameters (one paper).

The relationship between input data types and output data types in architectural design is also characterized by significant diversity with 61 unique combinations of input and output data types and contents. As a single input data type, images are often transformed into other images (74 papers, 45.97 %), while also being converted into 3d models (three), variables, and graphs (two papers each). The studies using variables as input focused on estimating the performance of designed results from the design parameters and conditions (17 papers, 10.56 %) and predicting optimized design parameters from the given conditions (eight papers). Text inputs mostly result in images (13 papers), highlighting their use in generating visual interpretations from input prompts, but also maintaining their format as text-based answers [23,25], relevant standard documents [138], and BIM objects representing object-oriented programming data schemes [22]. Graph inputs, which are mostly bubble diagrams, primarily maintain their format or are transformed into images of plan layouts. While other studies that generate 3D concept massing use images or variables, there is a study that uses partial 3D point clouds as features to complete 3D point clouds for concept mass generation [41].

Multimodal processing—where input data types such as image and variable (nine papers), image and text (eight papers), graph and variable, and image and graph (two papers each) are combined—enables generating more tailored design outcomes by providing cohesive guides

for design generation. These multimodal inputs predominantly result in images, in 19 out of 21 papers, reaffirming the prevailing focus on image generation. However, there are also instances where multimodal inputs lead to the creation of variables: one for predicting design parameters for parametric modeling from a bubble diagram and site conditions [99], and another for generating a building layout in graph format from a bubble diagram and design parameters [111]. There are four studies in which the output data type is classified as multimodal of images and variables. These studies commonly use two-step processing of images for object detection, identifying building components, symbols, and annotations, and semantic information extraction, attaining design parameters included in the annotations [114,116,117,137].

6.2. Distribution of input-output data type across AI model architectures

Fig. 12 illustrates the distribution of AI models across various input-output data types in architectural design. The heatmap is normalized across AI model architectures, with weighted totals annotated on the y-axis and the total number of papers for each input-output transformation on the x-axis.

FFNNs are primarily used on variable-to-variable transformations with 90.21 % of their applications focused on structured data processing. RNN is the only model architecture employed for text-to-variable transformations for classifying user sentiment from the given text [132], while also being used for an OCR application in image-to-multimodal (image with text) processing [114]. CNNs exhibit strong capacity in image processing with their use in image-to-image (55.88 %), image-to-variable (5.88 %), and image-to-multimodal (image and variable) transformations (20.59 %). Additionally, CNNs are the only architecture employed for image-to-3D model (11.76 %) [37,38] and multimodal (Image with variable)-to-variable transformations [12], reflecting their versatility in handling image inputs. GNNs focus primarily on graph processing, with 63.83 % of their usage dedicated to graph-to-graph transformations, followed by graph-to-image (12.77 %), multimodal (graph with variable)-to-graph (12.77 %), and image-to-graph (4.26 %). This distribution is consistent with GNNs' architecture, which is specifically designed to process graph-structured data. GANs dominate the image-to-image category (77.83 %), contributing a weighted sum of 56.17. Their diverse usage includes image-to-graph (0.46 %), variable-to-variable (0.69 %), text-to-image (0.92 %), and graph-to-image (3.46 %). GANs are also frequently used in multimodal data processing, in 13.86 % of their use, including images with variables, text [85,94], and graphs [17,66], as well as graphs with variables [99]. GANs are notably the only architecture employed for 3D model-to-3D model transformation (1.39 %) [41]. The encoder-decoder model focuses primarily on image applications, with 40.48 % of its usage in image-to-image, multimodal (image with variable)-to-image (21.43 %), and use as an OCR model in

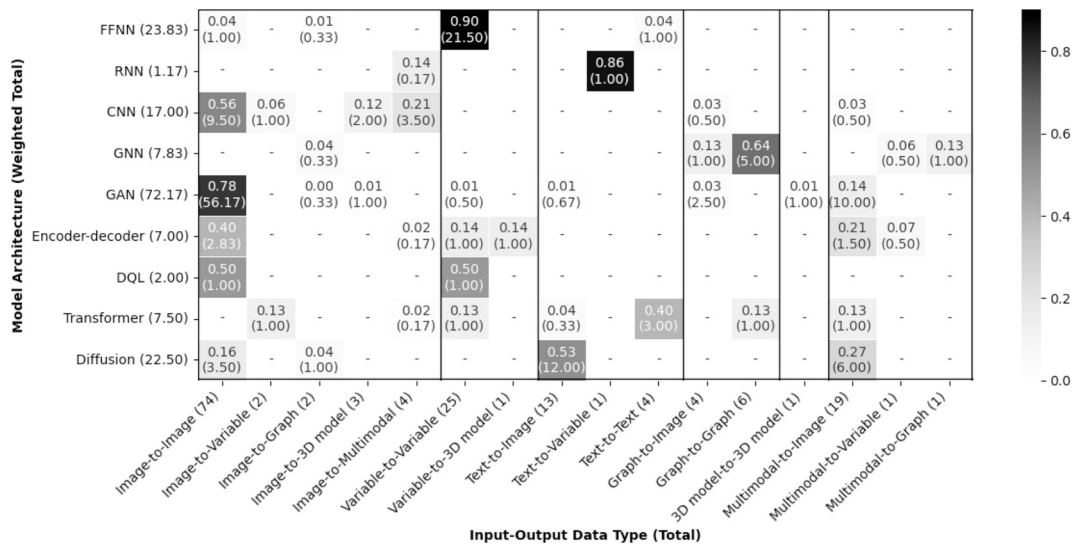


Fig. 12. Normalized distribution of input-output data type across AI model architectures.

image-to-multimodal transformation [114]. It is also employed for other data types for variable-to-3D models (14.29 %) and multimodal (graph with variable)-to-variable (7.14 %) transformations. DQL models are equally divided between image-to-image and variable-to-variable transformations (50 % each), reflecting their focus on decision-making tasks that span both visual and structured data inputs. Transformers demonstrate usage across various transformations. Text-to-text accounts for 40 % of their usage, followed by variable-to-variable, image-to-variable, graph-to-graph, and multimodal (image with text)-to-image (13.33 % each). It is also employed in text-to-image (4.44 %), and image-to-multimodal transformations (2.22 %). Lastly, diffusion models are particularly notable for their text-to-image transformations and

multimodal (mainly image with text)-to-image transformations accounting for 53.33 % and 26.67 % respectively, with a total of 80 %. This dominance highlights the capability of diffusion models in tailored image generation that reflects users' requests. Text-to-text and multimodal (mainly image and text)-to-image account for 46.15 % and 26.67 % respectively, emphasizing their growing importance in tailored image generation that reflects users' requests. Additionally, diffusion models are applied in image-to-image (15.56 %) and image-to-graph transformations (4.44 %). They are the only architecture employed for video generation tasks [19].

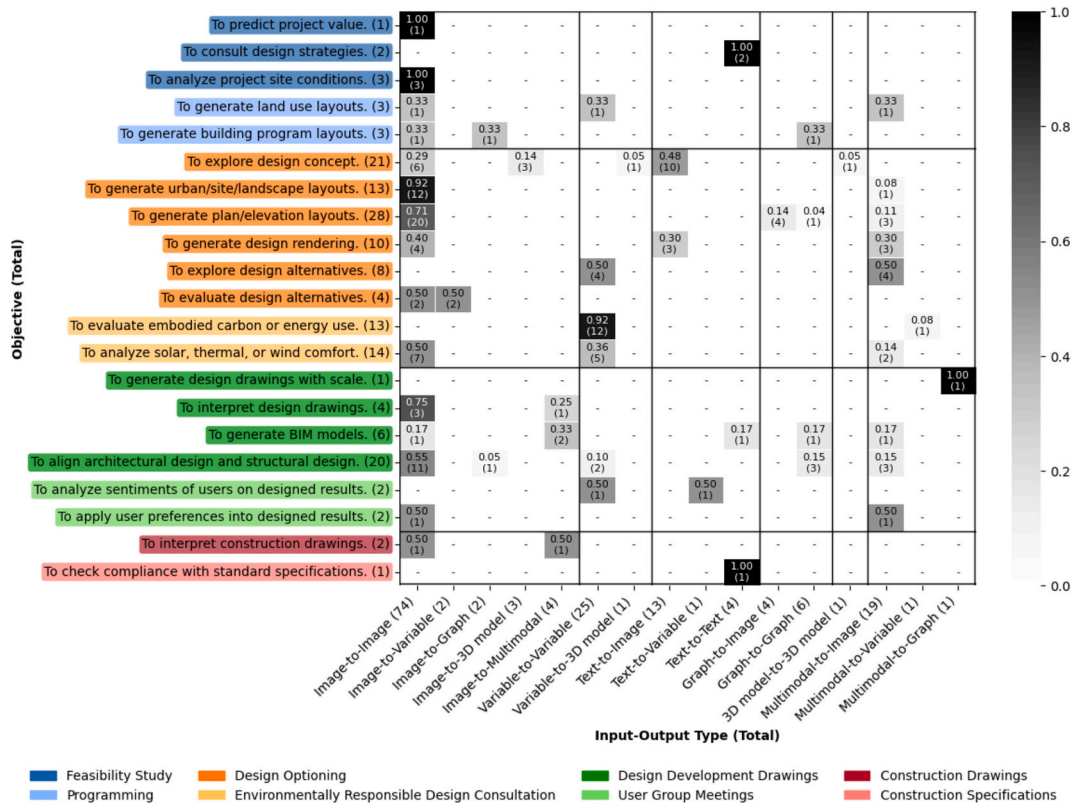


Fig. 13. Distribution of input-output data type across design objectives using AI.

6.3. Distribution of input-output data type across design objectives

Fig. 13 illustrates the distribution of input-output data types processed using AI across various architectural design objectives. The heatmap highlights which data types are utilized for each design task, offering insights into the types of transformations applied. To account for papers utilizing multiple formats, the data is normalized, and the total number of papers is annotated in parentheses.

In the *feasibility study* of the PD phase, the analysis revealed that the objectives of predicting project value, consulting design strategies, and analyzing project site conditions were approached using distinct data type transformations. Predicting project value [21] and analyzing project site conditions [26] both exclusively employed image-to-image transformations, highlighting the importance of identifying holistic conditions of the project site based on visual data that comprehends diverse factors. In contrast, consulting design strategies [23,25] utilized a text-to-text transformation, underscoring the importance of designers' intention-based planning during the strategic stage. For project *programming*, generating land use layouts involved a balanced approach, using image-to-image, variable-to-variable, and multimodal-to-image transformations, reflecting the diverse data types used in analyzing the project context, and determining their optimal use. On the other hand, generating building program layouts were demonstrated using image-to-image [31], image-to-graph [32], and graph-to-graph transformations [33], with all studies focusing on generating bubble diagrams.

In the *design optioning* of the SD phase, exploring design concepts (21 papers) employed various transformations, with a significant focus on text-to-image (10 papers, 47.62 %), along with image-to-image (6 papers, 28.57 %), image-to-3D model (three papers), variable-to-3D model, and 3D model-to-3D model formats (4.35 %, one paper each), which reflects the emphasis on translating the designer's conceptual idea into visualized results for design exploration. Generating urban/site/landscape layouts primarily relied on image-to-image transformations (12 papers, 92.31 %), with one use of multimodal-to-image. Similarly, generating plan/elevation layouts predominantly employed image-to-image (20 papers, 71.43 %), supplemented by graph-to-image (four papers), multimodal-to-image transformations (three papers), and graph-to-graph (one paper). Design rendering employed image-to-image (four papers), text-to-image (three papers), and multimodal-to-image transformations (three papers), illustrating the need to generate images in which specific styles or visualization strategies are to be reflected in the design results. For exploring design alternatives, multimodal-to-image and variable-to-variable transformations (50 %, four papers each) were used, highlighting the need to tailor the original design draft by providing a prompt along with the draft and to adjust design parameters to improve design outcomes (i.e., variable-to-variable). Studies that used AI to evaluate design alternatives included two papers each for predicting the estimated performance of the design alternatives as values (i.e., image-to-variable) or overlaying the values on the design alternatives (i.e., image-to-image). These versatile data types processed and generated in *design optioning* underscore a comprehensive approach required for exploring creativity yet aligning with the project scope.

For *environmentally responsible design consultation*, evaluating embodied carbon or energy use relies on variable-to-variable transformations (92.31 %, 12 papers), highlighting the focus on utilizing quantitative data such as design parameters and conditions to estimate the performance of the design results. This was complemented by a study that integrated a bubble diagram and factors such as meteorological conditions and seasonal variations (i.e., multimodal input) to predict energy use [99], suggesting that integrating conditions alongside design data enhances evaluation accuracy. Meanwhile, analyzing solar, thermal, or wind comfort involved a combination of image-to-image (50 %, seven papers), multimodal (image with variable)-to-image and -variable transformations (7.14 %, one paper each), and variable-to-variable transformations (35.71 %, five papers). This reflects the

frequent use of overlaid results as ground truth for training image generation models (i.e., surrogate models), following visualization strategies supported by environmental simulation tools used for training data generation.

For *design development drawings* of the DD phase, a study that utilizes multimodal (graph with variable)-to-graph for generating a building plan with scale [111]. This approach highlights the advantage of the graph format in representing contextual relationships between building components while providing robust data structures that encapsulate the complexities required for generating technical drawings. Interpreting design drawings primarily utilized image-to-image (three papers) and image-to-multimodal (image and text) transformations (one paper). Pizarro et al. [113] and Huang and Zheng [115] focused on segmenting building components and rooms within the drawing images. Schönfelder et al. [114] proposed a two-step approach for detecting text blocks in the drawings and extracting the design parameters (i.e., multimodal output) [114], which was further expanded in the studies aimed at generating BIM models. Similarly, Dodge et al. [112] proposed a three-step approach for parsing building objects and metadata together, using image segmentation, object detection, and OCR. Two other studies employed similar approaches to produce multimodal outputs of detected building components and design parameters for BIM model generation, using them as input for parametric modeling [116,117]. Kim et al. [120] proposed a method to add doors to the plan layout images and reflect them on BIM models (i.e., image-to-image). Additionally, Ghannad and Lee [119] conducted a study that utilized multimodal inputs (images and variables) to generate plan images and consequently produced a modularized BIM model. Their approach employed CoGAN to generate plan layouts and detect junctions using rule-based pixel-wise analysis, then converted them into BIM models parametrically represented in IFC format. Other papers for BIM model generation utilized text-to-text transformations to generate BIM objects in a structured text format based on user requests [22], and graph-to-graph transformations to create layout graphs where BIM housing modules are represented as nodes [118], reflecting efforts to address both the geometric and semantic information required in BIM models. Aligning architectural and structural design is the most extensively explored objective within the DD phase, exhibiting the widest range of transformations. These include straightforward methods such as image-to-image (55.56 %, 10 papers), graph-to-graph (16.67 %, three papers), and variable-to-variable (11.11 %, two papers), as well as more complex approaches, including image-to-graph, multimodal (image with variable)-to-image, and multimodal (image with text)-to-image (one paper each). This diversity highlights the significant research efforts dedicated to addressing this objective.

To support *user group meetings*, two studies were conducted—one utilizing users' textual interpretations of the building [132] and another employing EEG data collected as sequential values while users viewed the design results [133]—both aimed at classifying user sentiment as either positive or negative toward the design outcomes (i.e., variable). There are also efforts to incorporate user requests, particularly the public's ideas on urban and public facilities design, by using reference images [134] for image-to-image transformations or providing feedback for improvements [135] through multimodal (image with text)-to-image transformations.

In the CD phase, image-to-image transformations are employed for *construction drawing* interpretation to quantify building components from the construction BIM model view [137], and MEP drawings [137]. Interpreting *construction specifications* relies on text-to-text transformations to identify standard specifications that match the construction specifications, emphasizing the need to confirm design compliance based on the constraints and conditions required by standard documents or codes in this phase [137]. Both tasks, involving different data types and content, highlight the necessity of precision and the ability to handle complexity when interpreting construction documents.

7. Evaluation methods

This subsection presents a review of the evaluation types, methods, and metrics utilized in previous studies. Various combinations of evaluation types (four, including comparative evaluation, assessments by authors, third-party assessments, and machine-based assessments) and methods (three methods for each type) were employed in the studies. Additionally, 105 individual metrics were used, reflecting the multifaceted and complex nature of evaluating the performance of generative AI models in architectural design. We first examined the distribution of evaluation types and methods and then analyzed the distribution of evaluation methods across the input-output data types, comprehensively exploring the evaluation metrics utilized in each case.

7.1. Distribution of evaluation types and methods

Fig. 14 displays the evaluation types used for assessing generative AI in architectural design. The evaluation types for assessing AI models in architectural design reveal diverse yet skewed distributions. The most common evaluation type is comparative analysis, used in 98 of 161 papers (60.9 %), where ground truth data is applied to rigorously assess model performance. Author-led assessments are also prevalent, appearing in 55 papers (34.2 %), followed by third-party assessments (28 papers, 17.4 %) and machine-based assessments (14 papers, 8.7 %). Among the 161 papers, 134 (83.23 %) used a single evaluation type, while the other 27 papers employed hybrid approaches, where methods from different evaluation types were combined. Comparative evaluation was often a key component in these hybrid approaches, accounting for 21 of the 27 hybrid cases (77.8 %).

Fig. 15 illustrates the number of publications employing each evaluation method. The total count for each method exceeds the total number of papers because some studies used multiple methods from the same evaluation type. For example, a study using both criteria-based and numeric interpretation would count once under evaluation types but twice under evaluation methods. Dataset-based validation is the most frequently used method (63 papers, 39.13 %), underscoring the reliance on training data for performance assessment. Simulated results are employed in 27 papers to benchmark AI outputs (16.78 %), and expert comparisons are used in 14 papers to assess AI results against human-generated designs (8.70 %). In author-led evaluations, visual interpretation is the most common (42 papers, 26.09 %), followed by numeric interpretation (14 papers, 8.70 %) and criteria-based interpretation (5 papers, 3.11 %). For third-party assessments, surveys (16 papers, 13.22 %), expert evaluations (14 papers, 8.70 %), and the Turing

test (5 papers, 3.11 %) are employed. Machine-based assessments appear in 17 papers, with algorithm-based methods (7 papers, 4.35 % each) and unsupervised approaches (6 papers, 3.73 % each), followed by supervised approaches (4 papers, 2.48 %).

In summary, the frequent use of comparative evaluations—relying on datasets, simulations, and expert results—highlights the importance of establishing a reliable ground truth for evaluating AI-generated designs. Meanwhile, the prevalence of author and third-party assessments highlights the challenges in evaluating the quality of AI-generated designs without clear benchmarks. This underscores the need for evaluation methods to determine whether the generated designs align with practical knowledge and the designer's intent.

7.2. Distribution of evaluation method across AI model architectures

Fig. 16 illustrates the distribution of evaluation methods across various input-output data types. The heatmap is normalized across evaluation methods, with weighted totals annotated on the y-axis and the total number of papers for each input-output transformation on the x-axis.

The evaluation methods used to assess FFNNs, RNNs, and GNNs are relatively straightforward. For FFNNs, comparison with simulated results (62.94 %) and dataset validation (32.87 %) accounted for 95.81 % of the studies, indicating a strong reliance on comparative evaluation. RNNs are evaluated solely by employing dataset validation (100 %). Similarly, studies involving GNNs predominantly focused on dataset validation (78.72 %) and comparison with expert results (12.77 %), with a total of 91.59 % utilizing comparative evaluation. Notably, both FFNNs and GNNs are often used for processing singular data types—FFNNs for variable-to-variable tasks (90.21 %) and GNNs for graph-to-graph tasks (63.84 %). This unimodal nature likely simplifies the evaluation process for these models, making their assessments more straightforward.

Two studies utilizing DQLs assessed their methods using numeric interpretation and comparisons with simulated results (one each). Veloso and Krishnamurti [70] evaluated their multi-agent-based parametric design method by assessing the optimization rate on each design goal (i.e., numeric interpretation). Pan et al. [103] collected energy simulation results from DesignBuilder and used them as ground truth to evaluate their proposed multi-objective optimization method for improving building performances, including building energy consumption, carbon emission, and indoor thermal discomfort (i.e., simulated results).

Transformer models exhibited the highest proportion of comparison

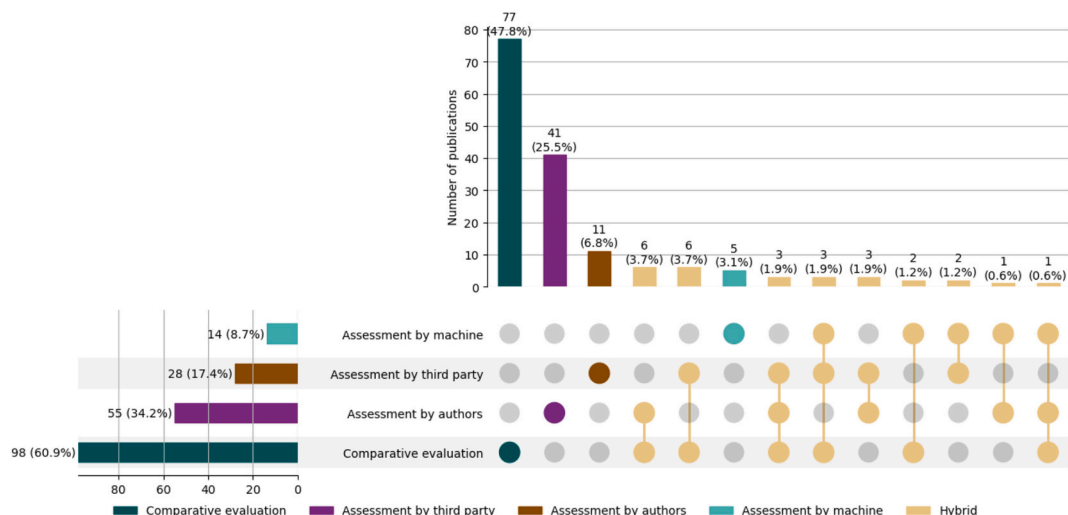


Fig. 14. Evaluation types utilized for assessing generative AI in architectural design.

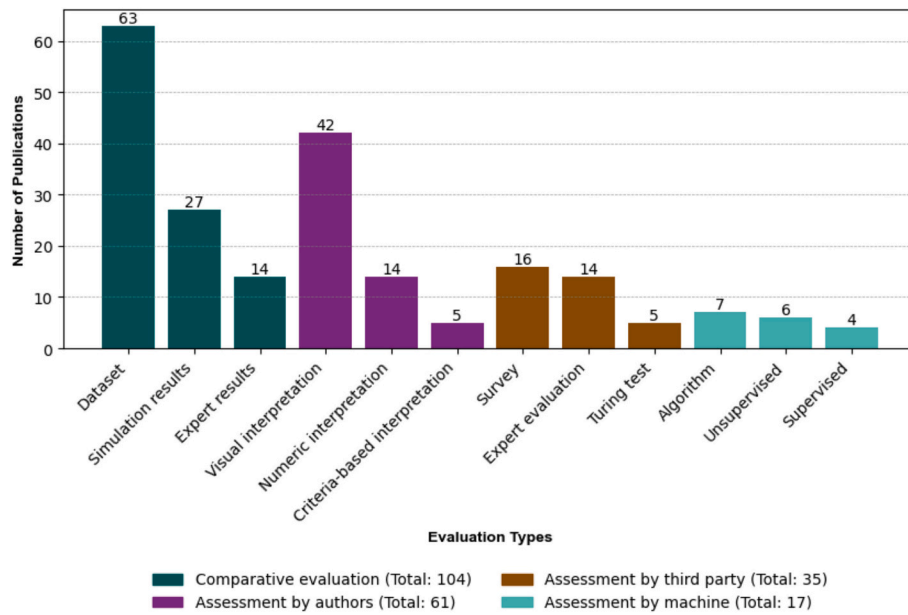


Fig. 15. Evaluation methods utilized for assessing generative AI in architectural design.

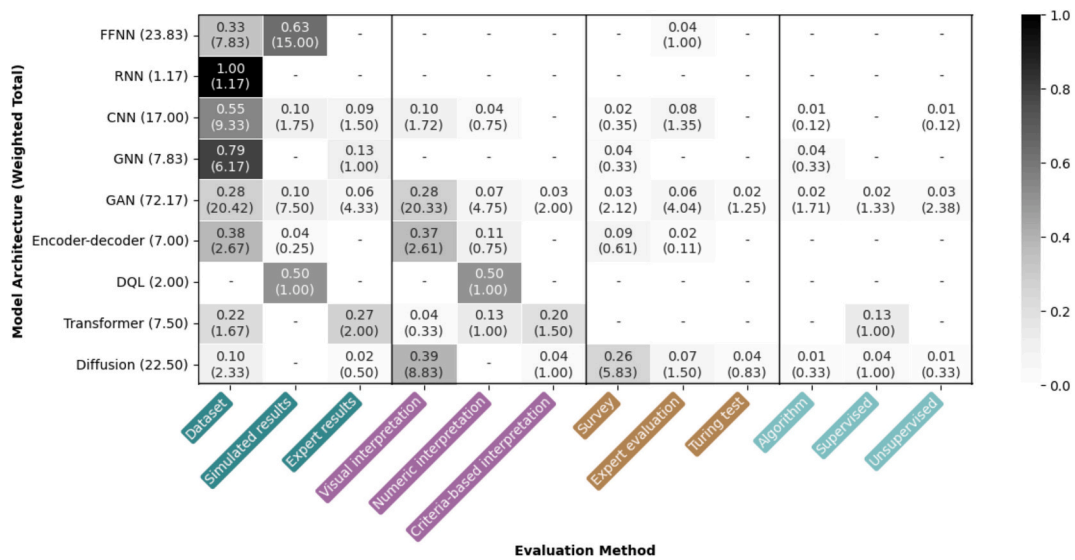


Fig. 16. Normalized distribution of evaluation methods across AI model architecture in architectural design.

with expert results (26.67 %), followed by dataset validation (22.22 %), criteria-based interpretation (20 %), numeric interpretation (13.33 %), and notable use of supervised machine evaluation (13.33 %). This distribution reflects their predominant use in natural language processing (NLP) tasks that require evaluating subtle nuances in text.

Image processing models tend to employ a diverse range of evaluation methods. CNNs, a more conventional vision model architecture, are often used for tasks such as object classification. Image processing models employ a diverse range of evaluation methods due to the varying nature of their tasks. CNNs, as a conventional vision model architecture, are often used for tasks such as object classification [120], object detection, instance segmentation [113,117], and semantic segmentation, where labeled ground truth is readily available. Consequently, CNNs showed the highest percentage of dataset validation (54.9 %), followed by simulated results (10.29 %), and expert results (8.82 %), with a total of 74.01 % using comparative evaluation against ground truth. Models used for image generation, such as GANs, encoder-decoder

models, and diffusion models, exhibited more variety in their evaluation methods. GANs were evaluated using comparative evaluation (39.28 %) and assessed by authors (39.7 %), machines (7.51 %), and third parties (5.23 %). Encoder-decoder models were evaluated through assessment by authors (48.01 %), comparative evaluation (41.67 %), and third parties (10.38 %). The significant use of author assessments, often involving visual interpretation without metrics, reflects the challenges of objectively evaluating the generative capacity of AI models. Diffusion models were primarily evaluated using visual interpretations by the authors (39.26 %) and survey (25.93 %), highlighting the subjective nature of assessing the generative capabilities of these models, especially when they are influenced by input prompts.

7.3. Distribution of evaluation method across design objectives

Fig. 17 illustrates the distribution of evaluation methods applied to various architectural design objectives, normalized across these

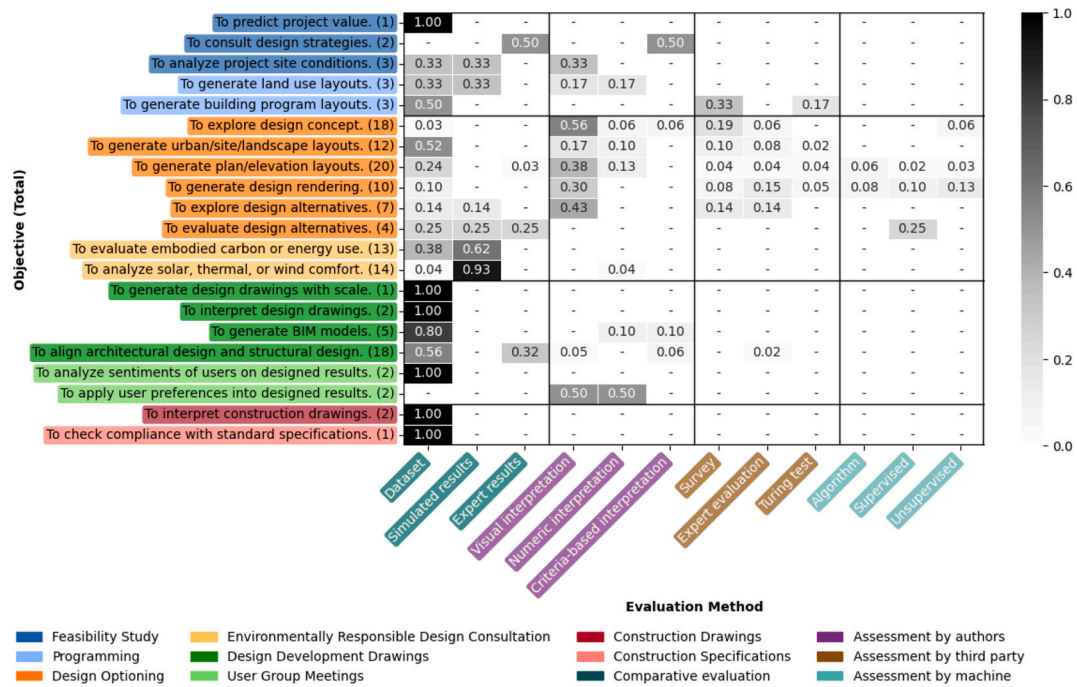


Fig. 17. Distribution of evaluation methods across architectural design objectives using AI.

objectives, and categorized by combinations of evaluation methods. The heatmap presents how different evaluation categories, such as metric-based, manual, and their combinations, are employed across a range of design objectives, with totals annotated on both the x- and y-axes. We further analyzed the evaluation metrics and performance indicators identified in the previous studies on each design objective in this subsection. The mathematical equations and description of the evaluation metrics are enlisted in Appendix B.

In the *feasibility study* of the PD phase (Table 5), the objective of predicting project value was exclusively evaluated using dataset-based validation (total count: 1), indicating a reliance on structured data to assess project viability. Specifically, Jiang et al. [21] compared the land value estimated by AI, overlaid on the urban map image, to its ground truth image using evaluation metrics such as mean absolute error (MAE), mean squared error (MSE), and structural similarity index measure (SSIM). Consulting design strategies (total count: 2), both

Table 5

Evaluation metrics and indicators to assess generative AI use during feasibility study of PD phase.

Design objective	Input-output data types	Evaluation objective	Evaluation metrics and indicators
To predict project value.	Image-to-image	To evaluate if the proposed method can accurately overlay estimate the land values on the given urban layout.	MAE, MSE, SSIM
To consult design strategies.	Text-to-text	To evaluate if the design strategy suggested by the AI properly addresses the design questions included in the input prompt.	ROUGE-1, ROUGE-2, ROUGE-L Suitability
To analyze project site conditions.	Image-to-image	To evaluate if the aerial view can be accurately converted into site or urban topography.	Precision, Recall, F1-score, IoU No metrics or indicators.
		To evaluate if the predicted urban environmental conditions accurately reflect the simulated results.	SSIM

conducted using text-to-text transformation, employed evaluation metrics of comparison with expert results and criteria-based interpretation (one each). Shelby and Da Silva [23] evaluated their RAG-based question-answering system by comparing the AI-suggested design strategies with expert-generated responses using recall-oriented understudy for gisting evaluation (ROUGE)-1, ROUGE-2, and ROUGE-L metrics. George et al. [36] assessed the suitability of the AI-suggested plants for landscape design by manually evaluating if the suggested plants conform to ecological, stylistic, functional, and aesthetic criteria to determine the suitability of the AI-suggested plants for given conditions. Project site conditions (total count: 3) were mostly assessed through comparative evaluation. Han et al. [26] used an image-to-image approach to convert aerial views into site or urban topography, evaluating the predicted land types and elevations over ground truth using precision, recall, F1-score, and intersection over union (IoU). A study conducted for predicting urban radiation from street images assessed whether the predicted urban environmental conditions accurately reflected simulated results, using SSIM as the metric [28].

In the *programming* of the PD phase, AI models generating land use layouts are evaluated primarily using comparative evaluation, including dataset validation and simulated results (one each), with one paper using numeric indicators and visual interpretation (Table 6). Wan et al. [29] evaluated if the predicted energy efficiency of the generated commercial district layout accurately reflects simulated results using RMSE, R^2 . Park et al. [14] utilized FAR, and BCR as indicators of AI-generated urban layout to assess conformance to the density constraints of the target site conditions. Liu et al. [30] assessed the similarity between the distribution of ground truth land use configuration and generated land use configuration using diverse similarity metrics such as average Kullback–Leibler divergence (KLD), average Jensen–Shannon divergence, average Hellinger distance, average cosine distance. For generating building program layouts (total count: 3), dataset validation was used (50 %), supplemented by surveys (33.33 %) and the Turing test (16.67 %). A spatial graph recommendation system developed by Park et al. [33] uses AI to recommend similar spatial graphs when given an input graph, trained to estimate GED. The researchers used precision and accuracy as evaluation metrics, reflecting how accurately the model estimates the actual GED between the input and recommended spatial

Table 6

Evaluation metrics and indicators to assess generative AI use during programming of PD phase.

Design objective	Input-output data types	Evaluation objective	Evaluation metrics and indicators
To generate land use layouts.	Image-to-image	To estimate the similarity between the distribution of ground truth land use configuration and generated land use configuration. To evaluate if the generated urban layout conforms to the target site and density parameters.	Average Kullback–Leibler Divergence, Average Jensen–Shannon Divergence, Average Hellinger Distance, Average Cosine Distance
	Variable-to-variable	To evaluate if the predicted energy efficiency of the generated commercial district layout accurately reflects simulated results.	Floor area ratio, Building cover ratio RMSE, R^2
To generate building program layouts.	Image-to-image	To evaluate the perceived realism of the generated program bubble diagrams. To estimate the similarity between the overall distribution of the generated and the real program bubble diagrams.	5-point Likert scale (Terrible to excellent) FID
	Image-to-graph	To estimate the authenticity of AI-generated program bubble diagrams perceived by humans. To estimate the perceived rationality of AI-generated program bubble diagrams.	Human accuracy 6-point Likert scale (Irrational to rational)
	Graph-to-graph	To evaluate the accuracy of predicted GED between the input and recommended program bubble diagram.	Precision, Accuracy

graphs. Other studies that generated spatial bubble diagrams based on given spatial boundaries employed Likert scales to enable third parties to evaluate the program layouts generated using AI. Additionally, Sun et al. [31] used Fréchet Inception distance (FID) to estimate the similarities between the overall distribution of the generated and ground truth bubble diagrams, while Su et al. [32] employed a Turing test, asking survey participants to distinguish AI-generated spatial diagrams from real spatial diagrams.

In the *design optioning* of the SD phase, exploring design concepts (total count: 21) leaned heavily on visual interpretation by authors without proposed metrics (57.14 %), as listed in Table 7. Most of the studies employed surveys (16.67 %), expert evaluation (4.76 %), and one study employed criteria-based assessment by authors, relying on subjective scoring as a performance measure. Paananen and Oppenlaender [48] utilized the creativity support index (CSI) to evaluate how the image generators support the participants' creative processes in terms of quality, novelty, and practicability. Other studies employed unsupervised machine evaluation (5.56 %), numeric interpretation (5.56 %), and dataset validation (2.78 %). Qian et al. [35,36] utilized FID to evaluate the similarities between the input and output sketches. This differs from previous applications of FID, which evaluated the distance between generated results and ground

Table 7

Evaluation metrics and indicators to assess generative AI for exploring design concepts during design optioning of SD phase.

Input-output data types	Evaluation objective	Evaluation metrics and indicators
Image-to-Image	To estimate the similarity between the original sketch and the AI-elaborated sketch. To evaluate how fast the training loss converges. To evaluate the reflection of conceptual initializers (Concept object, local music, and noise image) on design results.	FID Training loss No metrics
Text-to-Image	To evaluate how the image generators supported the participants' creative processes (Quality, novelty, and practicability). To estimate participants' responses to AI-generated design concept images. To estimate the preferences of the users on the design concept images. To evaluate if concept images generated using AIs utilize the proportions, colors, and ornaments of the given architectural style. To evaluate how the conditions and design parameters given as the input prompt are reflected on the concept images generated using AIs.	Creativity Support Index (CSI) Scoring (0–10) Scoring (Binary) Scoring (Ranking) Scoring (Binary) No metrics
Image-to-3D model	To evaluate how accurately the input sketch is converted into a 3D building mesh. To evaluate how efficiently the input sketch is converted into a 3D building To evaluate how well the generated 3D mesh reflects the features of a given concept image or training dataset.	Chamfer-L1 distance, Volumetric IoU, Average consistency score Computational cost No metrics
Variable-to-3D model	To evaluate how well an AI can imitate geometric operations.	No metrics
3D model-to-3D model	To evaluate the perceived usefulness of the proposed concept geometry generator in the design process.	No metrics

truth; in this study, FID is calculated to verify the similarity between the initial idea sketch provided by users and what was elaborated by the AI. Tono et al. [37] utilize Chamfer-L1 distance, volumetric IoU, and average consistency score to compare the 3D meshes generated from sketch inputs with the ground truth. They also evaluate the computational cost of the proposed method compared to other AI models to demonstrate its efficiency.

The capacity for generating urban, site, and landscape layouts was often comparatively analyzed using dataset validation (48.08 %) supported by an assessment of authors (32.69 %) and third parties (19.23 %) as enlisted in Table 8. In addition to MAE, MSE, and SSIM as previously identified for evaluating the AI-generated images over ground truth images, there are histogram distances, root MSE (RMSE), R^2 , peak signal-to-noise ratio (PSNR), and learned perceptual image patch similarity (LPIPS) employed in these studies. RMSE and R^2 are used to evaluate the pixel-level differences and the correlation between the predicted and actual images. PSNR is used to measure the ratio between the maximum possible power of a signal and the power of corrupting noise, indicating the quality of the reconstructed images. LPIPS is used to assess perceptual similarity between images by comparing deep features extracted from CNNs, providing a measure that aligns more closely with human visual perception. Studies have also used indicators such as floor space index (FSI), ground space index (GSI), and open space ratio

Table 8

Evaluation metrics and indicators to assess generative AI for generating urban/site/landscape layouts during design optioning of SD phase.

Input-output data types (contents)	Evaluation objective	Evaluation metrics and indicators
Image-to-image	To evaluate compliance of AI-generated urban layout with given design conditions	Floor Space Index (FSI), Ground Space Index (GSI), Number of floors (L = FSI/GSI), Open Space Ratio (OSR)
	To evaluate the diversity of the AI-generated layout.	Block number distance
	To estimate the authenticity of AI-generated landscape layout design results perceived by humans.	Human accuracy
	To estimate participants' responses to AI-generated landscape layout design results.	Scoring (0–10) Scoring (Binary)
	To understand the viability of AI-generated urban and landscape layout design results.	No metrics
	To estimate the similarity between the human- and AI-generated layout images.	Histogram distance SSIM PSNR MAE, MSE LPIPS RMSE, R ² KLD
	To evaluate the similarity between the design parameters of human- and AI-generated urban layouts.	
	To estimate the similarity between the human-generated layout and the AI-generated layout.	MAE, MSE, SSIM, PSNR
Multimodal (image with variable)-to-image		

(OSR) to evaluate the compliance of AI-generated urban layouts with given design conditions, and block number distances to evaluate the diversity of AI-generated urban layouts.

Table 9 enlists the evaluation metrics and indicators employed to assess AI models developed to generate plan and elevation layouts. For assessments conducted by authors (46.42 %), visual interpretation accounts for 32.14 %, and numeric interpretation for 10.71 %. While most visual interpretations are carried out without specific metrics [63], a few studies used principal component analysis (PCA) to visualize and interpret the features of AI-generated design results [17,78]. Para et al. [64] utilized topological statistics (S_t), element shape statistics (S_p), and alignment statistics (S_a), numerically evaluating the topology, dimension, and alignment of building elements within an AI-generated plan. Veloso et al. [70] utilized optimization rates as performance indicators to evaluate their method of generating building layouts considering multiple design goals. For comparative evaluation, previously identified metrics such as FID, SSIM, PSNR, precision, recall, and IoU are also used in evaluating AI-generated floor plans against ground truth. Additionally, Wang et al. [75] employed MSE and MAE to evaluate these floor plans. They also utilized normalized mutual information (NMI) to compare the ground-truth human-activity map, which guided the AI model in generating the plan layout, with the synthesized human-activity map derived from the generated plan. The assessment by third parties mainly utilizes the Likert scale, scoring, and Turing test. Similar to the Turing test, Zheng and Petzold [16] evaluated the perceived reasonableness of AI-generated floor plans against real floor plans through surveys. Similarly, Wang et al. [75] applied the Elo rating system to compare AI-generated plans with ground truth, assessing their relative quality across multiple questions by updating scores based on the outcomes. Additionally, various machine-based assessments are utilized. In algorithm-based approaches, GED is used to evaluate the similarity between input bubble diagrams and AI-generated plan layouts

Table 9

Evaluation metrics and indicators to assess generative AI for generating plan/elevation layouts during design optioning of SD phase.

Input-output data types (contents)	Evaluation objective	Evaluation metrics and indicators
Image-to-image	To estimate the similarity between the AI-generated plan layout and the ground truth.	FID SSIM KID
	To evaluate the diversity of the AI-generated results.	IS
	To compare the AI-generated plan layout against ground truth.	Precision, Recall, IoU NMI, MSE, MAE
	To evaluate the quality of the AI-generated layout image.	PSNR
	To compare and analyze the features of the images generated by the AI model.	PCA distribution
	To evaluate how optimized the plan layout design results are on each design goal.	Optimization rate
	To understand the spatial features of plan layouts generated using AI.	Room proportions, Visibility score
	To estimate the authenticity of AI-generated plan layout design results perceived by humans.	Human accuracy
	To evaluate spatial clarity and functional organization of AI-generated plan layout.	6-point Likert scale 5-point Likert scale
	To evaluate the superiority of results.	Scoring (Ranking), Scoring (Binary) Elo rating Selection rate Training loss
Graph-to-graph	To evaluate how fast the training loss converges.	No metrics
	To assess the logical placement of components.	
Graph-to-image	To evaluate the similarity between AI-generated plan layouts and ground truth.	FID
	To evaluate distribution of topology, dimension, and alignment of building elements within floor plan.	Topological statistics (S_t), Element shape statistics (S_p), Alignment statistics (S_a)
Multimodal (image with variable)-to-image	To evaluate the similarity between AI-generated plan layouts and ground truth.	FID
	To evaluate the perceived reasonableness of the floor plans generated using AI against real floor plans.	GED
Multimodal (image with graph)-to-image	To evaluate the perceived reasonableness of the floor plans generated using AI against real floor plans.	Selection rate
	To evaluate if AI-generated building plan layout can be practically utilized.	Scoring (0–10) Scoring (1–10)
Multimodal (image with graph)-to-image	To evaluate the similarity between AI-generated plan layouts and ground truth.	FID
	To analyze the features of the images generated by the AI model.	PCA distribution
Multimodal (image with graph)-to-image	To evaluate the similarity between AI-generated plan layouts and ground truth.	FID
	To evaluate if AI-generated images are classified as real or synthetic images.	Scoring (Binary)

(continued on next page)

Table 9 (continued)

Input-output data types (contents)	Evaluation objective	Evaluation metrics and indicators
	To evaluate the similarity between the input bubble diagram and plan layout generated using AI.	GED
	To evaluate the perceived reasonableness of the floor plans generated using AI against real floor plans.	Selection rate

[16]. Park et al. [15] employed DepthmapX to compute visibility scores to assess whether privacy requirements were met. In unsupervised approaches, Rodrigues and Duarte [20] used the Inception score (IS) to measure the variance in AI-generated floor plans. For supervised approaches, Aalaei et al. [17] trained an SVC on synthetic building layout images and used it as an evaluator to assess the perceived authenticity of AI-generated floor plans.

Table 10 enlists the evaluation metrics and indicators used to assess generative AI renderers. The evaluation of the rendering capacity of AI models involves a distribution between assessments by authors (30 %), third parties (28.33 %), and machines (31.67 %), with minimal use of comparative evaluations (10 %). All papers that employed author assessments descriptively interpreted how well AI-rendered images reflected user prompts, their functionality, and aesthetics. Zhou and Park collected paired datasets of rendered images and synthetic sketch images using an algorithm-based style transfer to train a Pix2Pix model, automating the generation of rendered images from sketches. This study is the only study utilizing comparative evaluation; however, it only interprets the difference between the AI-rendered image and its ground truth counterpart without employing evaluation metrics. Third-party

Table 10

Evaluation metrics and indicators to assess generative AI to generate design rendering during design optioning of SD phase.

Input-output data types (contents)	Evaluation objective	Evaluation metrics and indicators
Image-to-image	To evaluate the shape similarity between the AI-generated rendering graph and input sketch	FID
	To estimate the authenticity of AI-generated plan layout design results perceived by humans.	Human accuracy
	To evaluate the shape similarity between the AI-generated rendering graph against the input sketch and reference image	MS-SSIM
	To estimate the perceived style of AI-generated rendering.	Keyword frequency
	To evaluate clarity, diversity, and stylistic expression in the generated rendering.	No metrics
Text-to-image	To evaluate the functionality and aesthetics of AI-generated rendering or rendering assets.	Scoring (1–10) Scoring (Binary)
Multimodal (image and text)-to-image	To consistency of AI-generated interior video.	No metrics SSIM
	To evaluate the similarity between the AI-generated video and input image.	FID
	To evaluate the functionality and aesthetics of AI-generated videos.	Multiple choices
	To evaluate the similarity between the AI-generated video and the input prompt.	CLIP score
	To evaluate if the AI-generated image reflects the input prompt.	No metrics
	To evaluate the reflection of the requested style and image quality (Predicted by machine).	5-point Likert scale

assessments primarily involve scoring and the Turing test. Huang et al. [143] analyzed survey participants' reflections on rendered images, conducting a frequency analysis of keywords to determine if the rendered images reflected the designed architectural styles. Significant machine-based assessments were also employed to evaluate the rendering capabilities of AI. For algorithm-based approaches, SSIM and multiscale SSIM (MS-SSIM) were used to evaluate the structural similarities between input and output images [19,84]. Similarly, FID is used as an unsupervised approach to compare the similarities between input and output images [19,82,84]. In addition, Shao et al. [19] utilized the CLIP score to compare the similarities between the embedded prompt and the CLIP embedding of the generated rendered video frames as an unsupervised approach. For a supervised approach, Jo et al. [18] trained CNN models on human evaluations to assess whether the generated images aligned with the façades of the buildings (i.e., fidelity of domain), whether they satisfy high-quality visual criteria (i.e., image quality), and how effectively the regional design style is reflected (i.e., similarity to the applied region).

Table 11 enlists the metrics and indicators used to evaluate AI models for the generation and evaluation of design alternatives. AI

Table 11

Evaluation metrics and indicators to assess generative AI to generate and evaluate design alternatives during design optioning of the SD phase.

Design objective	Input-output data types (contents)	Evaluation objective	Evaluation metrics and indicators
To explore design alternatives.	Variable-to-variable	To compare the predicted optimized design parameters against ground truth. To evaluate how well the design parameters predicted by AI reflect the desired style. To evaluate how the AI-generated result optimizes material usage. To understand the distribution of the design space explored using AI.	R ² , MSE, RMSE, MAE, Precision, Recall, F1-score Accuracy Selection rate
	Multimodal (image with text)-to-image	To evaluate the aesthetic preferences of AI-generated alternatives against real design results. To evaluate how controlled the AI-generated images are.	Material use No metrics
	Multimodal (image with variable)-to-image	To analyze the shape distribution and generalization capabilities of the AI-generated design alternatives.	Selection rate No metrics
To evaluate design alternatives.	Image-to-image	To compare the AI-detected building components over ground truth. To compare the predicted possible route congestions against simulated results. To evaluate the visual comfort of parametrically generated design renderings. To estimate the similarity of AI-predicted aesthetic features between human evaluation.	AP, Recall, Mean IoU SSIM, PSNR Visual comfort index Accuracy
	Image-to-variable		

models for generating design alternatives, and visual interpretation by the authors are the most frequently used (43.75 %), followed by an equal distribution between surveys, expert evaluations, datasets, and simulated results (12.5 % each). Visual interpretation did not employ specific metrics. Danhaive and Mueller [92] evaluated the optimization of AI-generated design alternatives by comparing material usage. Third-party assessments focused on selection rates. Zhang et al. [144] presented participants with real and synthetic image imitations and asked for their aesthetic preferences. Zheng and Yuan [91] trained an MLP to predict design parameters for adjusting skyscrapers based on the styles of renowned architects. The authors then asked participating architects to select which architect's style was reflected in the skyscraper alternatives generated using the predicted design parameters. Two studies utilizing comparative evaluation trained their AI models for optimal design parameter exploration and validated their methods by comparing the predicted design parameters with those optimized in the dataset or through simulation. The studies focused on evaluating design alternatives, primarily utilizing comparative evaluation. One paper each focused on dataset validation, comparison with simulated results, and comparison with expert results. Moradi et al. [97] proposed a method for evaluating building plan layouts using CNN-based object detection models, which were evaluated against the ground truth using average precision (AP), recall, and mean IoU. Nourkojouri et al. [96] evaluated their model, trained to predict evacuation congestion, by comparing it against simulated results using SSIM and PSNR. Park et al. [98] trained a MobileNet model to automatically evaluate three aesthetic aspects of facade modules, with the performance of the subjective aesthetic evaluation assessed against expert evaluation results. Gui et al. [88] applied a different approach by using an AI model for design evaluation. They used a vision transformer to evaluate the visual comfort index of renderings generated through parametric modeling. However, they did not evaluate the performance of the vision transformer itself, as it was outside the scope of their study.

In the *environmentally responsible design consulting* of the SD phase, the performance of AI in evaluating embodied carbon, energy use, and analyzing solar, thermal, or wind comfort is predominantly assessed using comparative analysis (100 % and 96.43 %, respectively) as enlisted in Table 12. For evaluating AI models' accuracy in predicting embodied carbon and energy use, comparison with simulated results (61.54 %) is more frequently used than dataset validation (38.46 %), which often involves data collected from real-world cases. Although the methods vary across papers, the metrics used largely overlap, including R^2 , MAE, mean absolute percentage error (MAPE), MSE, and RMSE. Additionally, a study by Pomponi et al. [145] assessing the capability of AI to replace conventional FEM-based analysis of embodied carbon employed maximum error and explained variance scores as metrics. In addition, Feng et al. [10] used predicted global warming potential (GWP) as an indicator, comparing AI-generated GWP with Monte Carlo simulation results. The use of simulated results as ground truth is more prevalent in analyzing the environmental comfort of design results, with 92.86 % of studies adopting this approach. The metrics used are similar to those for evaluating the performance of embodied carbon and energy predictions, including R^2 , MAE, MAPE, MSE, and RMSE. Additionally, Wang et al. [108] used R-value, and Shen et al. [146] utilized mean absolute deviation (MAD). Several studies also employed SSIM and PSNR to compare predicted environmental conditions overlaid on the design with simulated environmental performances. Furthermore, Hou [106] utilized average detection accuracy (ADA) and successful detection probability (SDP) to evaluate whether the model accurately identified the problematic buildings within the residential block in terms of wind comfort.

In the *design development drawing* tasks of the DD phase (Table 13), only one study focused on generating design drawings to scale, which was evaluated using MSE and RMSE by comparing the vectored dimensions and scales between AI-generated floor layouts and the ground truth [111]. Studies conducted for interpreting design drawings,

Table 12

Evaluation metrics and indicators to assess generative AI use during environmentally responsible design consultation of SD phase.

Design objective	Input-output data types (contents)	Evaluation objective	Evaluation metrics and indicators
To evaluate embodied carbon or energy use.	Variable-to-variable	To compare the predicted embodied carbon or energy use against real data.	R^2 MAE MAPE MSE RMSE
		To compare the predicted embodied carbon or energy use against simulated results.	R^2 MAE MAPE MSE RMSE Max error, Explained variance score. Global Warming Potential (GWP) MSE
To analyze solar, thermal, or wind comfort.	Multimodal (graph with variable) Image-to-image	To compare the predicted energy use against real data.	R^2 MAE MSE RMSE SSIM PSNR ADA, SDP
		To compare the predicted solar, thermal, or wind performance against simulated results.	
	Variable-to-Variable	To evaluate if the model accurately determines the problematic buildings from the residential block.	Training loss
		To evaluate how fast the training loss converges.	
	Multimodal (image with variable)-to-image	To compare the predicted solar, thermal, or wind performance against simulated results.	R-value R^2 MAE MAD MSE RMSE
		To compare the predicted solar, thermal, or wind performance against simulated results.	SSIM MAE, MAPE, R^2 , MSE

commonly involving image detection, have commonly evaluated their methods using dataset validation. Pizarro utilized metrics such as IoU and mean IoU [113] to evaluate the performance of wall object segmentation. Similarly, Dodge et al. [112] utilized Jaccard Index (JI) to compare the detection against ground truth. Huang and Zheng [115] utilized activation possibilities, evaluating the likelihood of architectural areas being co-activated, reflecting spatial relationships during room semantic segmentation. Two studies conducted for the interpretation of design drawings [112,114] and two studies proposed methods to generate BIM models from drawings [116,117] utilized mean average precision (mAP), and weighted mAP (wmAP) which are often notated with numbers after '@' indicating the IoU threshold used for evaluation. For instance, $wmAP@0.5:0.05:0.95$ evaluates performance across multiple IoU thresholds, typically from 0.5 to 0.95 in increments of 0.05, providing a more comprehensive measure of the model's accuracy at different levels of overlap. In addition, a study by Phillip et al. [114] which conducted OCR to read text from the technical drawings, used the character error rate (CER) to evaluate the accuracy of text extraction. Lie et al. [118] who proposed a method utilizing BIM household modules to generate optimal layouts for high-rise residential buildings, evaluated the AI-generated layouts against the ground truth using R^2 , MAE, and MAPE. Kim et al. [120] assessed their method, which automatically

Table 13

Evaluation metrics and indicators to assess generative AI to generate and interpret design drawings and generate BIM models during DD phase.

Design objective	Input-output data types (contents)	Evaluation objective	Evaluation metrics and indicators
To generate drawings with scale.	Multimodal (graph with variable)-to-graph	To compare the difference of vectorized dimensions and scales between AI-generated floor layout and the ground truth.	MSE, RMSE
To interpret design drawings.	Image-to-image	To compare the detected symbols and components against ground truth. To evaluate the likelihood of architectural areas being co-activated, reflecting spatial relationships and learned design patterns in neural networks.	IoU Mean IoU mAP@0.5, Jaccard Index (JI) Activation possibility
	Image-to-multimodal (image and text)	To compare the detected objects against ground truth. To compare the predicted text string against ground truth.	wmAP@0.5;0.05;0.95 CER
To generate BIM models.	Image-to-image	To evaluate the detected or segmented building components against ground truth. To evaluate how accurately the AI places doors. To evaluate how accurately the AI classifies door types.	AP, wmAP mAP Accuracy Accuracy, Precision, Recall, F1-score
	Graph-to-graph	To compare the predicted BIM module layout against ground truth. To evaluate how fast the training loss converges.	R ² , MAE, MAPE Hubor loss
	Text-to-text	To estimate the extent to which the AI-generated design detail adheres to user requests. To estimate the extent to which the AI-generated design detail complies with engineering requirements.	Accuracy
	Multimodal (image and variable)-to-image	To evaluate the similarity between AI-generated and real building layout. To evaluate the practicability (connectivity, circulation, compactness, and space quality) of AI-generated building layout.	FID 3-point Likert scale

places the doors on the layouts, using accuracy, precision, recall, and F1-score as metrics. Ghannad and Lee [119] proposed a three-step method consisting of encoder-decoder-based building layout generation, joint detection through pixel-wise analysis, and parametric reconstruction of IFC models. They evaluated the performance of the AI model using the FID metric, similar to methods used for assessing AI-generated floor plans. Additionally, they assessed the practicality of the AI-generated BIM models by evaluating connectivity, circulation, compactness, and space quality through expert reviews. Jang et al. [22] proposed a framework for integrating LLMs into a natural language-based BIM authoring process. Accuracy was used to estimate the extent to which the AI-generated design detail adheres to user requests (i.e., criteria-based interpretation), and to estimate compliance with engineering requirements (i.e., numeric interpretation).

Aligning architectural design and structural design is primarily assessed using comparative evaluation (89.17 %), supported by author assessment (9.17 %) and third-party assessment (1.67 %). Common evaluation metrics used for comparing AI-generated results against ground truth datasets, such as R², IoU, accuracy, precision, recall, and F1-score, are also identified in these studies (Table 14). The evaluation metrics proposed by the authors are diverse yet focus on analyzing the similarity between AI-generated results and ground truth from real design cases [127], reflecting important engineering aspects in structural design [121]. Comparisons against expert results (34.17 %) focus on a more comprehensive understanding of AI-generated structural layouts, including design efficiency, material consumption, as well as structural features such as story drift ratios, torsional period ratios, and vibration periods [147].

Table 15 lists the evaluation metrics and performance indicators identified in studies conducted for the *user group meeting* during the DD phase. Two approaches proposed for analyzing user sentiment used

Table 14

Evaluation metrics and indicators to assess generative AI for aligning architectural design and structural design during design development drawings of DD phase.

Input-output data types (contents)	Evaluation objective	Evaluation metrics and indicators
Image-to-image	To compare AI-generated structural layout against ground truth in the dataset.	R ² IoU SIoU, WIoU, 1- η SWratio, SWratio _{GAN} , Score _{IoU} SIoU-M, S _{FloorA} , S _{IDR} Story drift ratio Material consumption Efficiency Torsional period ratio, Vibration period Scoring (1–5) Error percentage
Variable-to-variable	To evaluate the rationality and reasonability of AI-generated structural layout.	R ²
Graph-to-graph	To compare AI-generated structural layout against ground truth in the dataset.	Score _{DC} , IoU _{SW} , PN _{SWunion} Accuracy, F1-score, Recall, Precision IoU
Image-to-graph	To compare AI-generated structural layout against expert-generated structural layout.	IoU Precision, Recall Graph similarity Efficiency, Story drift ratio, Material consumption
Multimodal (image with text)	To compare AI-generated structural layout against expert-generated structural layout.	Score _{IoU} , FID
Multimodal (image with variable)	To compare AI-generated structural layout against ground truth in the dataset.	M _{Overall}

Table 15

Evaluation metrics and indicators to assess generative AI to support user group meetings of DD phase.

Design objective	Input-output data types (contents)	Evaluation objective	Evaluation metrics and indicators
To analyze the sentiments of users on designed results.	Text-to-variable	To compare the AI-predicted user sentiment from user response text against ground truth.	Accuracy
	Variable-to-variable	To compare the AI-predicted user sentiment from electroencephalography against ground truth.	Accuracy
To apply user preferences to designed results.	Image-to-image	To evaluate the diversity of AI-generated results in terms of quantity, density, and geometric features.	Number of buildings, Number of building components, BCR, FAR, Average height, Standard deviation of height, Average compactness, Standard deviation of compactness
	Multimodal (image with text)-to-image	To evaluate how well the AI-generated images of public parks reflect design conditions given by the public.	No metrics

accuracy as a metric, evaluating the AI-predicted user sentiment against the ground truth in the dataset [132,133]. Studies focused on reflecting user preferences in the design results were assessed by the authors. Quan [134] evaluated whether the AI-modified urban design reflects the topological features of the reference urban design including the number of buildings, number of building components, BCR, FAR, average building height, standard deviation of building heights, average compactness, and standard deviation of compactness. Guridi et al. [135] employed a visual interpretation of how the AI-generated images of public parks reflect design conditions given by the public although no specific metric was utilized.

The studies addressing tasks during the CD phase, including *interpreting construction drawings* and *estimating detailed costs from construction documents*, commonly used dataset validation (Table 16). Chowdhury and Moon [137] who proposed a method to automate the quantity take-off of building components through object detection evaluated the performance utilizing mAP@0.75, meaning the predicted bounding box

Table 16

Evaluation metrics and indicators to assess generative AI to interpret construction drawings and specifications in CD phase.

Design objective	Input-output data types (contents)	Evaluation objective	Evaluation metrics and indicators
To interpret construction drawings.	Image-to-image	To evaluate the detected building components against ground truth.	mAP@0.5 mAP@0.75 Accuracy, Recall, Precision, F1-score
To check compliance with standard specifications.	Text-to-text	To evaluate how relevant the standard specification retrieved by the AI is to the construction specification.	NDCG@1, NDCG@5, NDCG@10

must overlap the ground truth box by at least 75 %. Similarly, Jamieson et al. [136] utilized mAP@0.5, as well as accuracy, recall, precision, and F1-score to evaluate their models for detecting MEP symbols. Lee et al. [138], who proposed a method to automatically retrieve relevant standard specifications when a construction specification is given, utilized normalized discounted cumulative gain (NDCG). NDCG measures how well a ranked list matches an ideal ranking, taking both relevance and position into account, based on the number of items considered. For instance, NDCG@1 considers only the top-ranked item, evaluating how relevant and well-positioned it is, while NDCG@10 considers the top 10 items in the ranking, assessing the cumulative relevance and the positional discount for each of these items compared to an ideal ranking.

8. Discussion

This paper conducted a systematic review to provide a detailed understanding of the current state of how generative AI is transforming the roles and responsibilities of architects. The bibliometric and keyword analysis reveals the growing interest in employing generative AI in architectural design, focusing on early-stage design evaluation, the reflection of human ideas into design outcomes, and improving the efficiency of the design process. The answers to the research questions can be summarized as follows:

8.1. Reduced theory-application gaps

The overall trend suggests that the adoption of AI models in architectural design has increased significantly in recent years, with GANs leading the way, followed by a broader integration of other AI model architectures such as FFNNs, CNNs, GNNs, and encoder-decoder models. Although introduced later, the recent increase in the use of transformers and diffusion models with interactive text- and image-generation capabilities is gaining momentum, emphasizing the importance of reflecting users' design intent to the design results. The rapid increase in the number of publications underscores the growing interest in employing AI technologies in the architectural design generation.

The period between the emergence of the theory and the adoption of AI model architectures in architectural design decreased from 62 years (FFNN) to 2.5 years (Diffusion), resulting in a 96.09 % reduction in the theory-application gap. The fitted exponential decay curve ($y = 4807.37 \times \exp(-0.000217890 \times x) - 4062.33$) and linear regression ($y = -0.96 \times x + 736.42$) demonstrate a consistent decrease in theory-application gaps for AI adoption in architectural design. The exponential decay model achieves a slightly lower MSE of 511.46 compared to 538.01 for the linear regression model. However, the linear regression model provides a more interpretable representation of the steady reduction in adoption delays, underscoring the continued acceleration of AI adoption in the field.

8.2. Primary focus on early design tasks

Employing AI for the SD phase, where the exploration of design options and the selection of optimal design alternatives dominate, is the primary focus of previous studies (111 papers, 68.94 %). Especially, *design optioning* was most extensively studied (84 papers, 52.17 %), addressing design objectives including generating plan and elevation layouts (28 papers), exploring design concepts (21 papers), generating urban, site, and landscape layouts (13 papers), and generating design renderings (10 papers). These studies used variations of GANs such as Pix2Pix, CycleGAN, StyleGAN, FastGAN, and diffusion models such as Midjourney, Stable Diffusion, and DALL-E 2, along with custom domain diffusion models (e.g., FloorDiffusion [77] and FloorplanDiffusion [78]). *Environmentally responsible design consultation* follows with 27 papers (18.89 %), focusing on developing simplified prediction of solar, thermal, and wind comfort (14 papers) and embodied carbon or energy consumption (13 papers). These surrogate models are mainly built using

MLPs as FFNN-based approaches and using Pix2Pix as GAN-based approaches. While the DD phase follows next with 35 papers (21.74 %), previous studies mostly focused on aligning architectural design and structural design (31 papers) using diverse AI models from MLP, GraphSAGE, ChatGPT, as well as fine-tuned GANs (e.g., FrameGAN [148], StructGAN [123], StructTopo-GAN [122]) and diffusion models (e.g., Struct-Diffusion [126]). To interpret drawings and generate BIM models, CNN-based object detection models such as mask R-CNN, faster R-CNN, and diverse version of YOLO models [116] are primarily used.

Overall, generative AI is primarily utilized in the early design phase, focusing on exploring concept design exploration, generating design layouts, developing surrogate models for sustainability analysis, and rendering. While aligning architectural and structural designs is one of the most studied objectives in the later design phases, other complex tasks, such as generating construction drawings and specifications, are not identified in the selected literature.

8.3. Dominance of image-based planar design representations and potentials of other data types

Generative AI in architectural design utilizes five main data types—images, variables, text, graphs, and 3D models—with 30 distinct contents, with image format being dominant as input (85 papers, 52.8 %), output (110 papers, 68.32 %), and input-output (74 papers, 45.97 %). Images often represented planar design results such as boundaries, drawings, and layouts, as well as other design views such as perspectives, aerial views, and renderings. In addition, diverse contents such as local music using audio spectrograms, encoded voxels, and rendering video are represented using images. Variables are primarily used to input design parameters and conditions to attain predicted performance or optimized design parameters as output. Text data are mostly used as input prompts for tailored design generation, supporting interactive design processes. Graphs are mostly used to represent the contextual arrangement of design results, such as bubble program diagrams, drawings, and layouts. 3D models are used to represent concept masses in digital space. There are also multimodal data types utilized, including images with variables, text, and graphs, as well as graphs with variables. These distributions highlight the current emphasis on processing image-formatted design content while improving the processing of design semantics through diverse data types.

Each AI model architecture is tailored to process different input-output transformations. FFNNs are primarily used for variable-to-variable transformations, often paired with parametric modeling to generate design results or train surrogate models. RNNs, specifically LSTMs, are the only architecture used for user sentiment analysis from text data (i.e., text-to-variable). CNNs are typically used for detecting building components from images and performing OCR to extract semantics (i.e., image-to-multimodal), while GNNs mainly handle graph-to-graph transformations. GANs are applied to image generation tasks, such as semantic segmentation and style transfer. Similarly, encoder-decoder models are primarily used for image processing and multimodal (image with variable)-to-image generation. Transformers-based approaches such as GPT and RAG-enhanced BERT exhibited the potential to be used for design consultation [23,25], and interactive BIM modeling [22], whereas diffusion models excelled at generating images from prompts or multimodal inputs, such as images with prompts. This diversity in AI model architectures' capabilities underscores the importance of selecting the appropriate model architecture to achieve desired outcomes in architectural design.

While various early design tasks primarily involve generating images, several tasks, especially in the later phases, require different output types. Exploring design alternatives is conducted by predicting optimal design parameters given specific conditions (i.e., variable prediction). Evaluating design alternatives involves predicting desired building performance based on the given design parameters and conditions, and it is also applied to developing environmentally responsible

design strategies. Studies predicting user sentiments have also used variables as output. Text was the only output data type used for consulting design strategies, supporting interactive BIM-based design processes, and checking compliance of construction specifications with standard specifications. The only study that generated scaled drawings employed graph data to comprehend contextual relationships between building components and required numeric properties. Interpreting drawings and generating BIM models were achieved using image-to-multimodal (image and variable) formats by extracting geometric and semantic information from the drawings. These results emphasize the importance of employing various semantics to address later-phase design tasks, such as generating and interpreting drawings and ensuring compliance with design standards, as these tasks require complex yet robust semantics beyond simple visual representation.

8.4. Fit-for-purpose evaluation methods and needs for reflecting subjective factors

Comparative evaluation is the most commonly used method (98 papers, 60.9 %), supported by assessments from authors (55 papers, 34.2 %), third parties (28 papers, 17.4 %), and machines (14 papers, 8.7 %). Comparative evaluations utilize datasets (63 papers, 39.13 %), simulated results (27 papers, 16.78 %), and expert results (14 papers, 8.70 %) as ground truth. Visual interpretation is the primary method for author assessments (42 papers, 26.09 %). These methods, individually or in combination, are used to evaluate the performance of AI models in generating architectural designs.

Conventional AI model architectures, such as FFNNs, RNNs, CNNs, and GNNs, are easier to analyze objectively by comparing them to datasets or simulated results. In contrast, models with image generation capabilities, such as GANs and encoder-decoder models, often rely on author evaluations. Transformers and diffusion models, which reflect user intent but lack a clear ground truth, present challenges for comparative evaluation. These large models, pre-trained on extensive datasets, can generate highly customized design outputs with little additional training, complicating evaluation methods that often rely on authors, third parties, and machines.

Most studies in the DD and CD phases, and environmentally responsible design consultation in the SD phase, used comparative evaluation as the primary method for assessing AI models. However, studies covering the PD and SD phases employed diverse evaluation methods. This highlights the difficulty of evaluating early design phases, where there is no single optimal solution; design strategies reflect subjective preferences and require consideration of multifaceted goals.

Studies comparing ground truth images or variables with AI-generated results typically use metrics such as R^2 , MAE, MAPE, MAD, MSE, and RMSE, as well as performance measures such as accuracy, precision, recall, and F1-score. Some studies assess similarities between AI-generated outputs and ground truth using rule-based metrics like SSIM, MS-SSIM, GED, or unsupervised AI-based measures such as FID, IS score, and CLIP score. Overlap between ground truth and AI predictions is identified using JI, IoU, mean IoU, AP, mAP, and wAP. NDCG and ROUGE are also used to measure the similarity between AI-generated answers and desired answers. Studies that compare expert evaluations, particularly in aligning architectural and structural plans, often use efficiency, material consumption, and structural features such as story drift ratio, torsional period ratio, and vibration period. Additional indicators used to assess designs include GWP, BCR, FAR, average height, and number of components. These varied metrics (105 unique metrics within 161 papers) highlight the importance of selecting appropriate evaluation methods. While many studies use diverse metrics, a significant number rely solely on visual interpretation or subjective analysis (41 papers, 25.5 %), such as Likert scales or scoring, emphasizing the need to incorporate subjective aspects when evaluating design outcomes.

9. Conclusions

From a systematic review of 161 papers published between 2014 and 2024, this study confirms a rapid surge in generative AI adoption within architectural design, reducing the gap between theory and practice by roughly 96 % (from 62 to 2.5 years). Although traditionally dominated by feed-forward neural networks and convolutional architectures, more recent approaches—particularly diffusion models and transformers—are gaining momentum due to their ability to reflect user intent in real-time. AI applications remain heavily concentrated in the schematic design phase (about 69 %), with early-stage tasks like concept exploration and design optioning most frequently addressed, whereas later stages involving detailed documentation and construction receive comparatively less attention. Images still dominate as both input and output, but emerging studies highlight the value of variables, text, graphs, and 3D models in capturing more nuanced design semantics. Evaluation methods rely largely on comparative measures combined with subjective judgment, yet the field still lacks standardized benchmarks and domain-specific metrics—a gap that underscores the importance of establishing robust frameworks for validating AI-generated outcomes in architectural practice. Building on these findings, this review identifies promising directions to strengthen AI's role in architectural practice.

Need for AI-augmented design frameworks: Current AI-based architectural design methods often focus on integrating newly emerging models on a case-by-case basis, resulting in workflows that can quickly become obsolete. As AI technology continues to advance, practitioners are repeatedly forced to overhaul their processes to accommodate each new model or method. This approach is both inefficient and disruptive to ongoing design activities. Instead, there is a growing need for a more stable framework that establishes clear roles and interactions among design tasks, AI models, and data inputs. By defining how and where AI contributes to each stage of the process, such a framework allows practitioners to incrementally integrate new models without discarding the entire workflow. As enhanced AI models become available, they can simply replace or supplement existing components, ensuring that the overall design process evolves smoothly alongside advancements in AI. This not only reduces the burden of continual methodological revisions but also provides researchers with a consistent structure to evaluate, compare, and refine emerging AI solutions.

Expanding AI application in overall design phases: While extensive studies have focused on the SD phase, there has been limited attention to more labor-intensive phases such as DD and CD, which involve repetitive and time-consuming tasks [8]. Generating design and construction drawings has received minimal attention both in academic studies and practice, with only one study combining multimodal (graph with variables)-to-graph transformations and parametric modeling to generate scaled design drawings [111]. This process requires robust and diverse semantics within the design results—a quality often lacking in current AI applications. This deficiency is demonstrated by studies on the “floorplan effect” [63], where solely using images often fails to capture dimensional accuracy and detail. Studies that extract geometric features and semantic variables from drawings further emphasize the need to incorporate diverse information when representing design documents in the later design phases. Integrating AI into established tools such as parametric modeling and BIM offers a promising solution by combining AI's generative capabilities with the control provided by robust parametric algorithms [8]. Several commercial AI-based architectural design tools were not considered in this review due to the inaccessibility of the information on the specific AI models. However, they represent the growing trend of integrating AI into controllable robust systems. Additionally, the recent advent of LLMs shows the potential to support not only early decision-making in the PD phase but also throughout the entire design process, offering design strategies and data-driven recommendations that consider contextual domain knowledge [149].

Need for benchmark datasets and evaluation metrics: The absence of standardized benchmark datasets and evaluation metrics significantly impedes the advancement of AI technologies in architectural design. Unlike computer science domains with well-defined tasks and datasets, architectural design is inherently unique and context-dependent, making direct comparisons between methods challenging. Current metrics, often borrowed from other fields, fail to address critical architectural considerations such as spatial usability, dimensional accuracy, and contextual specificity. Moreover, the lack of domain-specific datasets has resulted in fragmented evaluation methods, with 105 unique metrics identified across 161 collected papers. Establishing standardized benchmark datasets and evaluation metrics is essential for enabling consistent comparisons and evaluations of AI models. This standardization would accelerate technological progress by allowing researchers to systematically assess and refine their approaches, fostering a more rapid and coherent development of AI augmentation to architectural practice. Several studies have begun addressing this need by publicly sharing domain-specific datasets [33,58,62,80,90,113,150]. These contributions are crucial as they provide the necessary foundation for training AI models on relevant data, ensuring that the resulting designs respect the diversity and specificity inherent in architectural practices. By facilitating meaningful comparisons and evaluations, standardized datasets and metrics will not only drive innovation but also improve the reliability and applicability of AI technologies in actual architectural design tasks.

Enhancing human-in-the-loop control of AI-generated designs: The recent advent of transformers and diffusion models has enabled architects to use textual prompts to generate or modify design outcomes. However, a common difficulty identified in the previous studies is the continuous generation of unexpected results, requiring iteration of trial and error to fully reflect the architects' design intentions [48,53,87]. To address this problem and improve human-in-the-loop control, two possible future efforts are suggested. The first emphasizes fostering designers' abilities to articulate design intents more precisely. Architectural education should enable students to discern subtle nuances of terminology describing desired design results. Educators should focus on teaching detailed and concise expert vocabulary needed to communicate envisioned design concepts to generative AI models [48]. While this technique, known as prompt engineering, can be effective in retrieving desired outputs from AI models, it can also pose a technical barrier to applying AI in design practice. Therefore, the second suggestion is to develop methods that enhance human-in-the-loop control by enabling machines to better understand the designers' intent. This could involve creating more intuitive interfaces or algorithms that interpret designers' inputs more effectively, reducing the reliance on precise textual prompts. Enhancing human-in-the-loop control over AI-generated architectural designs must be a key focus in future studies to overcome these challenges.

Complementing these forward-looking recommendations, this review paper offers several key contributions. First, it quantitatively reveals the trends in AI adoption within the architectural design domain. Although it is generally recognized and experienced that AI adoption is accelerating—particularly through the extensive use of image-generation tools during early design phases—the precise figures have remained unclear. By presenting specific quantitative values (e.g., a 96.06 % reduction in the gap between theory and application, a 68.94 % usage rate in the SD phase, and a dominant use of images at 68.32 %), this paper provides concrete understanding on how AI is being introduced and applied. Second, it clarifies which types of AI models are used for particular design tasks and, conversely, provides an overview of which tasks currently benefit from AI adoption. For industry professionals, these findings serve as a roadmap of AI technologies applicable to real-world design workflows. Simultaneously, academic researchers can use these insights to identify research gaps and set priorities for future investigations. Third, this study presents a comprehensive use case illustrating how these models and their performance on

design tasks can be evaluated under varying conditions. Architectural design requires both high levels of expertise and creativity, making objective evaluation especially challenging. By systematically analyzing how different data types—such as images, variables, text, graphs, and 3D models—are used as inputs for AI models during each design phase, this review clarifies which evaluation metrics are employed to validate AI-generated outputs. These insights serve as a practical resource for both researchers and practitioners seeking to implement or assess AI-driven design processes, providing guidance on data handling, model selection, and performance evaluation. Finally, building on the key requirements and limitations identified in earlier research, we propose a set of future opportunities aimed at bridging these gaps and driving more robust AI adoption in architectural design.

Despite these contributions, certain limitations should be acknowledged. First, there are scope and timeliness constraints regarding the dataset. This study meticulously selected research papers on AI applications in architectural design from 2014 to 2024 through keyword, citation, and reference searches. However, as indicated by both the exponential and linear models, the adoption of AI in architectural design continues to accelerate at a pace that can outstrip publication timelines. Rapid developments in AI may have already taken place during the preparation and review of this manuscript, potentially rendering some findings outdated. Moreover, because most of the papers (148, or 91.93 %) were published within the last five years, it is challenging to identify truly influential publications, increasing the likelihood that some significant studies may have been overlooked.

In light of the shifts that AI is bringing to the field of architecture, this review underscores the need for a proactive, adaptive approach to integrating these technologies into design workflows. While AI has proven capable of enhancing creativity, optimizing design processes, and improving efficiency, it also raises important questions regarding sustainability, confined use in the early design phase, generalization, and lack of controls. A key priority moving forward will be the development of AI frameworks that streamline with architect's design practice, evolving in tandem with technological advancements. As AI is expected to continue transforming architectural practice, addressing

these challenges will be essential for ensuring that the technology serves not only as a tool for efficiency but also as a catalyst for creativity and innovation in design.

CRediT authorship contribution statement

Suhyung Jang: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Hyunsung Roh:** Writing – review & editing, Validation, Methodology, Data curation, Conceptualization. **Ghang Lee:** Writing – review & editing, Supervision, Project administration, Methodology, Funding acquisition, Conceptualization.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used OpenAI's tool, ChatGPT in order to proofread the manuscript. After using this tool, the authors reviewed and edited the content as needed and took full responsibility for the content of the publication.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Ghang Lee reports financial support was provided by Korea Agency for Infrastructure Technology Advancement. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.autcon.2025.106174>.

Appendix B. Evaluation metrics

Evaluation metrics with descriptions and their mathematical equations.

Metric	Description	Mathematical equation
$\eta_{SWratio}$	Correction coefficient for overall quantity of shear walls.	$1 - \left \frac{SWratio_{GAN} - SWratio_{target}}{SWratio_{GAN}} \right $
ADA (average detection accuracy)	Proportion of correctly detected buildings in each individual sample, averaged to reflect detection accuracy of test dataset.	$avg(ADA_i)$
AP (average precision)	Average measure of precision across recall levels, often used in ranking tasks.	$\int_0^1 PRdR$
mAP (mean Average Precision)	Mean of Average Precision scores for multiple classes or tasks.	$\frac{\sum_{i=1}^C AP_i}{C}$
wMAP (weighted mean Average Precision)	Weighted version of mAP, accounting for class imbalances.	$\sum_{i=1}^C \frac{AP_i * N_i}{M}$

Where:

- $SWratio_{GAN}$ refers to the shear wall ratios of the GAN designs
- $SWratio_{target}$ refers to the shear wall ratios of the target designs

Where:

- ADA_i : the proportion of correctly predicted buildings in the synthetic image of the i -th test sample

Where:

- R : The value of recall
- P : The value of precision
- C : The total number of classes
- AP_i : The average precision corresponding to the objects of class i
- N_i : The number of objects in the i -th class
- M : The total number of objects

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Metric	Description	Mathematical equation	
Accuracy	Ratio of correctly predicted instances to total number of instances, measuring overall correctness.	$\frac{TP + TN}{TP + TN + FP + FN}$	Where: - TP (True Positive): Correct predictions for positive class - TN (True Negative): Correct predictions for the negative class - FP (False Positive): Incorrect predictions where the model mistakenly identifies a negative as positive - FN (False Negative): Incorrect predictions where the model mistakenly identifies a positive as negative
Precision	Ratio of correctly predicted positive instances to the total predicted positives, reflecting quality of positive predictions.	$\frac{TP}{TP + FP}$	
Recall	Ratio of correctly predicted positive instances to total predicted positives, reflecting quality of positive predictions.	$\frac{TP}{TP + FN}$	
F1-score	Harmonic mean of precision and recall, providing a balance between the two, especially when they are in conflict.	$2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$	
Activation possibility	Likelihood that a certain element or neuron is activated, often reflecting its contribution to a specific outcome in a neural network.	$\frac{\text{Number of activations}}{\text{Total number of instances}}$	–
Average Cosine Distance	Average measure of similarity between two vectors based on the cosine of the angle between them.	$\frac{\sum_{j=1}^5 w_j \bullet \cos(Y_j, \hat{Y}_j)}{\sum_{j=1}^5 w_j}$	Where: - w_j : the weight of the green level j - Y_j : the distribution of original land-use configurations - $\hat{Y}_j$: the distribution of generated land-use configurations
Average Hellinger Distance (HD)	Average measure of the maximum distance between points on two surfaces or curves.	$\frac{\sum_{j=1}^5 w_j \bullet HD(Y_j, \hat{Y}_j)}{\sum_{j=1}^5 w_j}$	
Average Jensen-Shannon (JS) Divergence	Average symmetrized version of KL Divergence, measuring similarity between two probability distributions.	$\frac{\sum_{j=1}^5 w_j \bullet JS(Y_j, \hat{Y}_j)}{\sum_{j=1}^5 w_j}$	
Average Kullback-Leibler (KL) Divergence	Average measure of how one probability distribution diverges from a second, reference distribution.	$\frac{\sum_{j=1}^5 w_j \bullet KL(Y_j, \hat{Y}_j)}{\sum_{j=1}^5 w_j}$	
Average consistency score	Metric exploiting higher-order information by considering the surface normals of the meshes; represents mean absolute dot product between the normal of one mesh and normals at the nearest neighbors in the ground truth mesh.	$\frac{1}{N} \sum_{i=1}^N n_i \bullet n_i^{gt} $	Where: - n_i : the normal vector at the i -th point of the reconstructed mesh - n_i^{gt} : the normal vector at the nearest of that point in the ground truth mesh - N : the total number of points considered Where: - S : the necessary number of substitutions - D : the number of deletions - I : the number of insertions - N_{gt} : the total number of characters in the ground truth string Where: - c, v : the feature vectors output by the CLIP encoder for textual description - $\cos(c, v)$: the similarity between c and v calculated using the cosine function - w : the weight used to adjust the impact of similarity
CER (character error rate)	Metric used in text recognition to measure percentage of characters incorrectly predicted.	$\frac{S + D + I}{N_{gt}}$	Where: - c, v : the feature vectors output by the CLIP encoder for textual description - $\cos(c, v)$: the similarity between c and v calculated using the cosine function - w : the weight used to adjust the impact of similarity
CLIP score	Metric that measures alignment between images and text based on a multimodal model called CLIP.	$\text{Score}(c, v) = w \times \max(\cos(c, v), 0)$	Where: - c, v : the feature vectors output by the CLIP encoder for textual description - $\cos(c, v)$: the similarity between c and v calculated using the cosine function - w : the weight used to adjust the impact of similarity
Chamfer-L1 distance	Mean of accuracy and completeness metric.	Mean (accuracy, completeness metric)	Where: - accuracy: mean distance of points on y_{recon} to their nearest neighbors on the $\hat{y}_{recon}$ - completeness metric: distance of points from y_{recon} to $\hat{y}_{recon}$
Elo rating	System to calculate and update player ratings after a match, reflecting their relative skill levels based on expected and actual outcomes.	$R' = R + K(O - E)$	Where: - R' : New rating - R : Current rating - K : K-factor (sensitivity to rating changes, e.g., 20 or 40) - O : Actual match outcome (1 for win, 0.5 for draw, 0 for loss) - E : Expected outcome, calculated as $\frac{1}{1 + 10^{(R_{opponent} - R)/400}}$
Explained variance score	Metric that measures proportion of variance in dependent variable that is predictable from independent variables, indicating how well a model captures variability in data.	$\text{explained variance}(y, \hat{y}) = 1 - \frac{\text{Var}(y - \hat{y})}{\text{Var}(y)}$	Where: - $\text{Var}(y - \hat{y})$: the variance of prediction errors - $\text{Var}(y)$: the variance of actual values
FID (Frechet inception distance)	Metric to assess quality of generated images by comparing their feature distributions to real images.	$d^2 = \ \bar{u}_1 - \bar{u}_2\ ^2 + \text{Tr}(C_1 + C_2 - 2\sqrt{C_1 \times C_2})$	Where: - $\bar{u}_1, \bar{u}_2$: the element-wise vector mean of the real and generated images - C_1, C_2 : the covariance matrices for the real and generated feature vectors - $\ \bar{u}_1 - \bar{u}_2\ ^2$: the sum squared difference between the two mean vectors - Tr : the trace linear algebra operation
FSI (floor space index)	Ratio of a building's total floor area to size of plot of land on which it is built.	$\frac{F}{A}$	Where: - F : the total area of the buildings

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Metric	Description	Mathematical equation	
GSI (ground space index)	Ratio of a building's footprint to plot area, indicating construction density.	$\frac{B}{A}$	- B: the total area of the building footprints - A: the site area
L	Number of floors.	$\frac{FSI}{GSI}$	
OSR (open space ratio)	Ratio of open, unbuilt space to built-up area in a plot.	$\frac{1 - GSI}{FSI}$	
Floor area ratio (FAR)	Ratio of total floor space constructed divided by the lot area.	$\frac{Constructed\ floor\ space}{Lot\ area}$	-
Graph editing distance (GED)	Metric that measures similarity (or dissimilarity) between two graphs by counting the minimum number of edits needed to transform one graph into another.	$GED(g_1, g_2) = \min_{(e_1, \dots, e_k) \in \rho(g_1, g_2)} \sum_{i=1}^k c(e_i)$	Where: - $\rho(g_1, g_2)$: the set of edit paths transforming g_1 into g_2 - $c(e)$: the cost of each graph edit operation e
Graph similarity	Measure of how similar two graphs are in terms of structure or attributes.	$S(g_1, g_2) = \frac{GED(g_1, g_2)}{ V_1 + V_2 + E_1 + E_2 }$	Where: - V: vertices - E: edges
Histogram distance	Measure of the pixel distribution differences between two images.	$\sqrt{1 - \sum_i \sqrt{h_1(i)h_2(i)}}$	Where: - $h_1(i), h_2(i)$: the frequency of RGB value i in the histograms of GAN-generated layouts, respectively.
Huber loss	Loss function used in regression that is less sensitive to outliers than squared error.	$L_\delta(a) = \begin{cases} \frac{1}{2}a^2 & \text{for } a \leq \delta, \\ \delta \left(a - \frac{1}{2}\delta \right) & \text{otherwise} \end{cases}$	Where: - a : refers to the residuals - δ : a threshold parameter that controls the point at which the loss function transitions from a quadratic function to a linear function
Inception score (IS)	Measure of diversity and quality of generated images based on a pretrained Inception model.	$IS = \exp(E_x[D_{KL}(p(y x) p(y))])$	Where: - x : a generated image - $p(y x)$: the conditional label distribution predicted by the inception model for image x - $p(y)$: the marginal distribution over all images - $D_{KL}(p(y x) p(y))$: the Kullback-Leibler divergence, measuring how different the conditional distribution $p(y x)$ is from the marginal distribution $p(y)$
Intersection over Union (IoU)	Metric for evaluating overlap between predicted and ground truth regions in object detection.	$\frac{A \cap B}{A \cup B}$	Where: - A: generated output - B: original result
IoU_{SW}	Similarity of shear wall layout based on IoU.	$\frac{PN_{SW_{inter}}}{PN_{SW_{union}}}$	Where: - $PN_{SW_{union}} = PN_{SW_g} + PN_{SW_{GNN}} - PN_{SW_{inter}}$ - $PN_{SW_g}, PN_{SW_{GNN}}$: the shear wall pixel numbers of the engineer designed and GNN-designed shear wall layouts, respectively
Jaccard Index (JI)	Measures similarity between two sets by comparing their intersection to their union; often used in segmentation or classification tasks.	$\frac{ A \cap B }{ A \cup B }$	Where: - A: Predicted set - B: Ground truth set
Kernel inception distance (KID)	Metric similar to FID but based on polynomial kernel differences, used for evaluating generative models.	$MMD(p, q) = E_{x, x' \sim p}[K(x, x')] + E_{x, x' \sim q}[K(x, x')] - 2E_{x \sim p, x' \sim q}[K(x, x')]$	Where: - $E_{x, x' \sim p}[K(x, x')]$: the expected difference between 2 images from the image set p - $E_{x, x' \sim q}[K(x, x')]$: the expected difference between 2 images from the image set q - $E_{x \sim p, x' \sim q}[K(x, x')]$: the average difference between 2 images from the image set p and q , respectively
Kullback-Leibler divergence (KLD)	Measure of how much probability distribution of design attribute of synthesized layouts differs from that of real-world layouts.	$KLD(P Q) = \sum_{x \in X} P(x) \log \left(\frac{P(x)}{Q(x)} \right)$	Where: - $P(x)$: real-world layouts - $Q(x)$: synthesized layouts - X: the probability space
Learned perceptual image patch similarity (LPIPS)	Measure of perceptual similarity between images based on deep learning features.	$LPIPS = \sum_l \frac{1}{H_l W_l} \sum_{h, w} \ w^l \odot (\hat{y}_{hw}^l - \hat{y}_{ohw}^l)\ _2^2$	Where: - H_l, W_l : the height and width of the feature maps at layer l - $\hat{y}_{hw}^l, \hat{y}_{ohw}^l$: the feature activations of image - w^l : a learned weight applied to the differences at each layer
Mean absolute deviation (MAD)	Average of absolute deviations from the mean, indicating data variability.	$\frac{1}{N} \times \sum_{i=1}^N y_i - \bar{y} $	Where: - N: the number of observations - y_i : the i th observed value calculated by FEM - $\bar{y}$: the i th estimated value predicted by the ML models

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Metric	Description	Mathematical equation	
Mean absolute error (MAE)	Statistical metric used to measure accuracy of a model by evaluating average magnitude of errors between predicted values and actual values.	$MAE = \frac{1}{2} \sum_{i=1}^n \hat{y}_i - y_i $	
Mean absolute percentage error (MAPE)	Metric used to measure accuracy of a forecasting model by calculating percentage difference between actual and predicted values.	$MAPE = \frac{100}{n} \sum_{i=1}^n \left \frac{y_i - \hat{y}_i}{y_i} \right $	Where: - n : the number of data points - $\hat{y}_i$: the predicted value for the i -th data point - y_i : the true value for the i -th data point
Mean squared error (MSE)	Metric that measures average of squares of differences between actual values and predicted values.	$\frac{1}{n} \sum_{i=1}^n (\hat{y}_i - y_i)^2$	
Max error	Metric that measures maximum difference between predicted values and the actual values in a dataset.	$\max_{i=1} y_i - \hat{y}_i $	
Multiscale structural similarity index measure (MS-SSIM)	Measure of structural similarity as an approximation index between inputs and outputs	$M = M(A, G) \times 0.5 + M(B, G) \times 0.5$	Where: - A and B: input image - G: the fused image - when evaluating: A (the input sketch), B (the reference image)
$M_{overall}$	Comprehensive gauge to evaluate generated shear wall layout schemes by considering weights of VGG19, ScoreIoU, and MSE.	$W_{vgg19}M_{vgg19} + W_{scoreIoU}M_{scoreIoU} + W_{mse}M_{mse}$	Where: - W: the weights attributed to vgg19, scoreIoU, and mse - $M_{vgg19} = softmax(logits)_{label}$ - $softmax(x_i) = \frac{e^{x_i}}{\sum_{j=1}^K e^{x_j}}$ - $M_{mse} = 1 - \frac{sum(abs(S_{eng} - S_{gen}))}{sum(S_{eng})}$
Mean IoU	Average of IoU across all classes or instances in a dataset.	$\frac{1}{C} \sum_{c=1}^C IoU_c$	Where: - c: A class c - C: Total number of classes - IoUc: IoU for class c
Normalized discounted cumulative gain (NDCG)	Discounted cumulative gain (DCG) of the actual ranking normalized by the DCG of the ideal ranking NDCG@# refers to computing NDCG only up to position # in a ranked list. This is useful when you only care about the top # results (e.g., NDCG@10 would evaluate the quality of ranking only for the top 10 results).	$\frac{DCG}{IDCG}$	Where: - DCG = $\sum_{i=1}^N \frac{rel_i}{\log_2(i+1)}$ - IDCG = $\sum_{i=1}^N \frac{rel_i^{opt}}{\log_2(i+1)}$ - i: the position of the result - rel_i : the actual relevance score for the result at position i - rel_i^{opt} : the ideal relevance score for the result at position i
Normalized Mutual Information (NMI)	Metric to measure similarity between two clustering results, considering both shared and exclusive information in clustering assignments.	$NMI = \frac{2 \times I(X; Y)}{H(X) + H(Y)}$	Where: - I(X;Y): Mutual information between variables X and Y - H(X): Entropy of X - H(T): Entropy of Y
Peak signal-to-noise ratio (PSNR)	Metric to measure quality of a signal, often in context of image and video compression.	$10 \log \left(\frac{MaxValue^2}{MSE} \right)$	Where: - MSE: the mean squared error between two images - MaxValue: the maximum value the image pixels can take
R-value	Correlation between predicted output and target output.	$r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2} \sqrt{\sum (y_i - \bar{y})^2}}$	Where: - x_i : the predicted output - $\bar{x}$: the mean of the predicted output - y_i : the target output - $\bar{y}$: the mean of the target output
Root mean squared error (RMSE)	Metric for evaluating accuracy of a regression model.	$\sqrt{\frac{1}{N} \times \sum_{i=1}^N (y_i - \hat{y}_i)^2}$	Where: - n: the number of data points - y_i : the actual value for the i -th data point - $\hat{y}_i$: the predicted value for the i -th data point
Recall-oriented understudy for gisting evaluation (ROUGE)	ROUGE-1	Metric to measure overlap of unigrams between system-generated summary and reference summary.	$\frac{\text{Number of overlapping unigrams}}{\text{Total number of unigrams in reference summary}}$
	ROUGE-2	Metric to measure overlap of bigrams between system-generated summary and reference summary.	$\frac{\text{Number of overlapping bigrams}}{\text{Total number of bigrams in reference summary}}$
	ROUGE-L	Metric to measure longest common subsequence (LCS) between system-generated summary and reference summary.	$\frac{LCS(\text{reference, generated})}{\text{Number of words in reference s}}$
R-squared (R^2)	Statistical measure used to assess how well a regression model fits the data.	$1 - \frac{SS_{regression}}{SS_{total}}$	Where: - $SS_{residual}$: the sum of the squared differences between the actual values y_i and the predicted values $\hat{y}_i$

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Metric	Description	Mathematical equation	
Successful detection probability (SDP)	Proportion of samples with 100 % detection accuracy, which reflects probability of successful detection of test dataset.	$\frac{num(n_p == n_r)}{num_{total}}$	- SS_{total}: the sum of the squared differences between the actual values y_i and the mean of the actual values $\bar{y}$ - $SS_{residual} = \sum_{i=1}^n (y_i - \hat{y}_i)^2$ - $SS_{total} = \sum_{i=1}^n (y_i - \bar{y})^2$ Where: - n_p, n_r: the predicted and the real numbers of problematic buildings in a sample.
SIoU	IoU of shear walls, representing similarity between shear wall layouts designed by the AI and engineers.	$\frac{A_{SW_{inter}}}{A_{SW_{union}}}$	–
SSIM (structural similarity index measure)	Measure of similarity between two images considering luminance, contrast, and structure.	$SSIM(x, y) = [l(x, y)]^\alpha \cdot [c(x, y)]^\beta \cdot [s(x, y)]^\gamma$	Where: - $l(x, y)$: luminance - $c(x, y)$: contrast - $s(x, y)$: structure - α, β, γ: parameter (0–1)
SW_{ratio}	Shear wall ratio.	$\frac{A_{swall}}{A_{swall} + A_{inwall}}$	Where: - A_{swall}: the shear wall area - A_{inwall}: the infill wall area
S_{FloorA}	Supported ratio under the vertical load.	$1 - \frac{A_{minus}}{A_{floor}}$	Where: - A_{minus}: the floor area that is not supported by the shear wall - A_{floor}: the total area of the floor
S_{IDR}	Consistency of inter-story drifts of the structures designed by the GAN and engineer.	$1 - \left(\left 1 - \frac{\theta_{GAN,X}}{\theta_{ENG,X}} \right + \left 1 - \frac{\theta_{GAN,Y}}{\theta_{ENG,Y}} \right \right) / 2$	Where: - $\theta_{GAN,X}$ and $\theta_{GAN,Y}$: the maximum inter-story drifts of the GAN's design in the X and Y directions, respectively - $\theta_{ENG,X}$ and $\theta_{ENG,Y}$: the maximum inter-story drifts of the engineer's design in the X and Y directions, respectively
S_{IoU-M}	Modified planar layout consistency indicator.	$\eta_{DiffSwall} \frac{A_{inter}}{A_{union}}$	Where: - $\eta_{DiffSwall} = 1 - \frac{ L_{GAN} - L_{ENG} }{L_{ENG}}$, where L_{GAN}, L_{ENG} are the total lengths of the shear walls designed by the GAN and the engineer, respectively. - A_{inter}, A_{union}: the intersection and the union areas of the shear walls designed by the GAN and engineer, respectively.
$Score_{DC}$	Similarity of shear wall layout distribution and shear wall ratio, calculated by multiplying weight coefficients of shear wall ratio.	$IoU_{SW} \times w_1 \times w_2$	Where: - $w_1 = 1 - \frac{ SW_{gt1} - SW_{pre1} }{SW_{gt1}}$ - $w_2 = 1 - \frac{ SW_{gt2} - SW_{pre2} }{SW_{gt2}}$ - $SW_{gt1} = PN_{SW_{gt1}} / PN_{wall}$ - $SW_{gt2} = PN_{SW_{gt2}} / PN_{wall}$ - $SW_{pre1} = PN_{SW_{pre1}} / PN_{wall}$ - $SW_{pre2} = PN_{SW_{pre2}} / PN_{wall}$ - SW_{gt1}, SW_{gt2}: the shear wall ratios in the two orthogonal directions of the engineer's design results - SW_{pre1}, SW_{pre2}: the shear wall ratios in the two orthogonal directions of the GNN model's design results - $PN_{SW_{gt1}}, PN_{SW_{gt2}}$: the pixel number of the shear walls in the two orthogonal directions of the engineer's design result - $PN_{SW_{pre1}}, PN_{SW_{pre2}}$: the pixel number of the shear walls in the two orthogonal directions of the GNN model's design result - PN_{wall}: the pixel number of the PSW in the shear wall structure
WIoU	Weighted IoU	$\sum_{i=0}^k \frac{w_i p_{ii}}{\sum_{j=0}^k p_{ij} + \sum_{j=0}^k p_{ji} - p_{ii}}$	Where: - $(k + 1)$: the number of the class - p_{ij}: the pixel that belongs to class i but is predicted to be in class j

Data availability

Data will be made available on request.

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